भौतिक विज्ञान वन लाइनर

Unit 1: Units, Measurement, and Measuring Instruments

1. What is a Light Year a unit of?
   - Distance (Astronomical distance).
2. Which is the largest unit for measuring astronomical distances?
   - Parsec (1 Parsec = 3.26 Light Years).
3. What is measured in Cusec?
   - Flow of water.
4. What is the unit for measuring the thickness of the ozone layer in the atmosphere?
   - Dobson.
5. By whom is SONAR used?
   - By Navigators (to find the depth of the sea or submerged objects).
6. Which instrument measures the intensity of sound?
   - Audiometer.
7. Which instrument is used to measure high temperatures (like the Sun's heat)?
   - Pyrometer.
8. Which instrument measures atmospheric pressure?
   - Barometer.
9. A sudden fall in the Barometer reading is an indication of what?
   - Stormy weather.
10. By which instrument is the relative density or purity of milk measured?
    - Lactometer.
11. Which instrument is used to measure the humidity of the air?
    - Hygrometer.
12. What is the Richter Scale used to measure?
    - Intensity of an earthquake.
13. What is the unit of 'Power'?
    - Watt.
14. With which instrument is Electric Current measured?
    - Ammeter.
15. What is the name of the lie-detecting instrument?

- Polygraph.

16. What is the instrument used to measure Blood Pressure called?
    - Sphygmomanometer.

Unit 2: Mechanics (Force, Energy, Gravitation, Friction)

17. Which of Newton's laws is also called the 'Law of Inertia'?
    - First Law.
18. On which principle do washing machines and cream separators work?
    - Centrifugal Force.
19. 'It is harder to walk on ice than on a road', why?
    - Because there is less friction on ice compared to the road.
20. As soon as a train starts moving, the passengers sitting inside lean backwards. What is the reason for this?
    - Inertia of Rest.
21. Which energy is present in winds?
    - Only Kinetic Energy.
22. Who propounded the Universal Law of Gravitation?
    - Newton.
23. What is the value of Escape Velocity from the Earth's surface?
    - 11.2 km/second.
24. If Earth's gravitational force suddenly disappears, what will happen?
    - The weight of the object will become zero, but the Mass will remain the same.
25. Why do Pendulum Clocks become slow in summer?
    - Due to the increase in the length of the pendulum (Time period increases).
26. What is the height of a Geo-stationary Satellite from Earth?
    - Approximately 36,000 km.
27. Where is the weight of an object maximum on Earth?
    - At the Poles.

Unit 3: Properties of Matter (Surface Tension, Archimedes, Viscosity)

28. What is the reason for the spherical shape of a raindrop?
    - Surface Tension.

29. Oil spreads on the surface of water, why?
    - Because the surface tension of oil is less than that of water.
30. Due to what reason does kerosene rise up in the wick?
    - Due to Capillarity.
31. How is the pressure inside a soap bubble compared to atmospheric pressure?
    - It is higher.
32. At what temperature is the density of water maximum?
    - At 4°C.
33. When a ship enters the sea from a river, what happens?
    - It rises a little (because the density of sea water is higher).
34. Why does ink start leaking from a pen while traveling in an airplane?
    - Due to a decrease in atmospheric pressure.
35. What is the reason for clouds floating in the atmosphere?
    - Their low Density.

Unit 4: Heat & Thermodynamics

36. What is the normal temperature of the human body?
    - 37°C or 98.6°F.
37. At what temperature are the values of Celsius and Fahrenheit scales equal?
    - -40° (Minus 40 degrees).
38. What is the effect on the Boiling Point of water when salt is added to it?
    - The Boiling Point increases.
39. Which metal is the best conductor of heat?
    - Silver.
40. To keep a beverage at the same temperature for some time in a thermos bottle, whose layer is applied to its walls?
    - Silver layer (Silver Plating).
41. What is the main use of sweat?
    - To regulate body temperature.
42. Why does food cook faster in a pressure cooker?
    - The boiling point of water increases due to high pressure.

Unit 5: Sound & Wave

43. What type of waves are sound waves?
    - Longitudinal mechanical waves.
44. In which medium is the speed of sound maximum?
    - In Steel (Metal/Solids).
45. Why can't two people hear each other on the surface of the Moon?
    - Because there is no atmosphere there.
46. What is considered the standard sound level (noise limit) for humans?
    - 60 Decibels (dB).
47. What is the frequency of Ultrasonic Waves?
    - More than 20,000 Hertz (Hz).
48. Which waves do bats use to fly in the dark?
    - Ultrasonic waves.
49. What should be the minimum distance between the reflecting surface and the listener to hear a clear Echo?
    - 17 meters (approximately).

Unit 6: Light (Reflection, Refraction, Eye)

50. What is the nature of light radiations?
    - Like both Wave and Particle (Dual Nature).
51. What is the Speed of Light in a vacuum?
    - 3×108 meters/second.
52. Why does the sky appear blue?
    - Due to Scattering of light.
53. Why does the Sun appear red at sunrise and sunset?
    - Because the scattering of red color is the least.
54. Which mirror is used as a 'Rear view mirror' in vehicles?
    - Convex Mirror.
55. Which mirror is used for shaving?
    - Concave Mirror.
56. Which lens is used to correct Myopia (Nearsightedness)?
    - Concave Lens.

57. Which lens is used to correct Hypermetropia (Farsightedness)?
    - Convex Lens.
58. What is the reason for the formation of a Mirage?
    - Total Internal Reflection of light.
59. On which principle does Optical Fiber work?
    - Total Internal Reflection.
60. Why does a diamond appear shiny?
    - Due to Total Internal Reflection.
61. An air bubble in water will act exactly like a—?
    - Concave Lens.
62. What kind of image is formed on the retina of the human eye?
    - Real and Inverted.
63. What is the reason for the formation of a Rainbow?
    - Reflection, Refraction, and Dispersion of light.
64. Why is red light used for danger signals?
    - Because its scattering is the least (it is visible from a long distance).
65. Which are the Primary Colors?
    - Red, Green, and Blue.
66. What does the color of a star indicate?
    - Its Temperature.
67. Which technology is used to transmit audio signals in Television (TV) broadcasting?
    - Frequency Modulation (FM).

Unit 7: Electricity & Magnetism

68. What is the filament of an electric bulb made of?
    - Tungsten.
69. Which gas is filled inside an electric bulb?
    - Argon (or Nitrogen).
70. What is the source of electrical energy in an artificial satellite?
    - Solar Cells.
71. What is a Transformer used for?
    - To step up or step down (Up or Down) AC voltage.

72. Fuse wires used in homes are made of which alloy?
    - Tin and Lead – they have a low melting point.
73. Which device converts electrical energy into mechanical energy?
    - Electric Motor.
74. Who converts mechanical energy into electrical energy?
    - Dynamo.
75. The filament of a 'Halogen Lamp' is made of which alloy?
    - Tungsten and Sodium.
76. What is the full form of CFL?
    - Compact Fluorescent Lamp.
77. Ohm is the unit of what?
    - Electrical Resistance.
78. What is the cause of Earth's magnetic field?
    - Circulating currents inside the Core.
79. What is a mobile charger?
    - A Step-down Transformer.

Unit 8: Modern Physics, Electronics & Miscellaneous

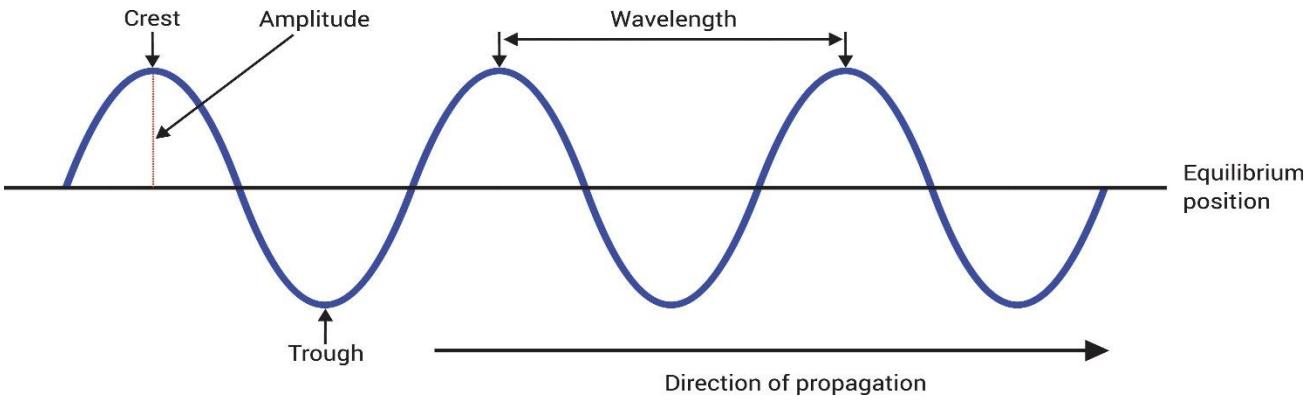
80. On which principle is the Atom Bomb based?
    - Nuclear Fission.
81. On which principle does the Hydrogen Bomb work?
    - Nuclear Fusion.
82. What is the source of the Sun's energy?
    - Nuclear Fusion.
83. Which nuclear fuel is available in abundance in India?
    - Thorium.
84. What is used as a Moderator in a nuclear reactor?
    - Heavy Water (D2O) and Graphite.
85. Curie is the unit of what?
    - Radioactivity.
86. What is RADAR primarily used for?
    - To detect the presence and position of objects using radio waves.

87. In what is the LASER beam used?
    - Eye Surgery and Barcode scanning.
88. What are the IC chips used in computers made of?
    - Silicon.
89. Which is the most used material for making transistors?
    - Silicon.
90. Which type of electromagnetic radiation is used for the remote control of television?
    - Infrared radiation.
91. What is the color of the 'Black Box' of an airplane?
    - Orange.
92. For which discovery did Albert Einstein receive the Nobel Prize?
    - For Photoelectric Effect.
93. Who invented the X-Ray?
    - Roentgen.
94. What is called the 'God Particle'?
    - Higgs Boson.
95. What does the color of the sky appear to be to an astronaut?
    - Black.
96. By what is the acidity of rain (Acid Rain) measured?
    - By pH meter.
97. How many colors are there in the Sun's rays?
    - 7 colors.
98. For which effect was C.V. Raman given the Nobel Prize?
    - For Raman Effect (Scattering of light).
99. When is National Science Day celebrated?
    - 28 February.
100. What is the value of one Kilowatt-hour (kWh) in Joules? * 3.6×106 Joules.

JOB MANTRA BY R D SIR Taiyari से Selection तक

भौतिक विज्ञान के महत्वपूर्ण प्रश्न जो पूर्व की परीक्षाओं में आये है-

1. What is a Light Year a unit of measurement for? — Distance traveled by light in a vacuum in one year.
2. One Horse Power is approximately equal to how many Watts? — 746 Watts.
3. The S.I. unit of Resistance is? — Ohm.
4. What is the commercial unit of electric power? — Kilowatt-hour (kWh).
5. What kind of quantity is Work? — Scalar quantity.
6. The SI unit of Stress is? — Pascal.
7. The SI unit of Volume is? — Cubic meter (m3).
8. The SI unit of Time is? — Second.
9. The SI unit of Magnetic Induction is? — Tesla.
10. The sound intensity of a musical instrument is measured in? — Decibels (dB).
11. The name of the unit for measuring frequency is? — Hertz (Hz).
12. Supersonic speed is measured by? — Mach number.
13. The unit of Force in the S.I. system is? — Newton.
14. The unit of Atmospheric Pressure is? — Bar.
15. The unit of Work and Energy is? — Joule.
16. The SI unit of Dynamic Viscosity is? — Pascal-second.
17. The unit of Power is? — Joule/second (or Watt).
18. The unit of Pressure is? — Pascal.
19. The S.I. unit of Electric Current is? — Ampere.
20. The definition of the CGS system of measurement units is? — Centimeter, Gram, Second.
21. 1 Watt-hour is equal to? — 3600 Joules.
22. 1 Volt is equal to? — 1 Joule per Coulomb.
23. Work, Kinetic Energy, and Potential Energy are what kind of quantities? — Scalar quantities.
24. The physical quantity whose dimension is [ML2T−2] is? — Work done.
25. The unit of Wavelength is? — Meter.



26. The instrument which is always connected in series in a circuit and measures electric current is called? — Ammeter.
27. The dimension of the physical quantity Power is? — [ML2T−3].
28. The name of the unit for measuring food energy is? — Calorie.
29. Represents the units of the same physical quantity? — Joule and Calorie (both for Energy/Heat).
30. The correct statement regarding Electric Potential is? — It is a scalar quantity.
31. 'Ringelmann Scale' is used to measure the density of? — Smoke.
32. One "Barrel" of oil is approximately? — 159 Liters.
33. When a rock is brought from the surface of the Moon to Earth, then? — Its weight will change but mass will not.
34. What kind of quantity is Displacement? — Vector quantity.
35. The unit of measurement which when multiplied by 0.39 gives 'Inch'? — Centimeter.
36. Electron Volt is the unit of? — Energy.
37. Megawatt is the unit of measuring electricity which is? — Generated (Production capacity).
38. The value of one Kilowatt-hour is? — 3.6×106 J.
39. The speed of airplanes and ships is displayed in 'Knots'. A speed of 100 Knots will be? — Equal to 115 miles per hour.
40. Wind speed is measured by? — Anemometer.
41. Which is not a unit of distance? — Square mile (It is a unit of Area).
42. Vector magnitude is? — Force.
43. Which physical quantity has the same SI unit as Work? — Energy.

44. In the CGS (Centimeter-Gram-Second) system, the unit of Magnetic Intensity is? — Oersted.
45. A non-SI unit called 'nit', used to measure the abundance of light intensity is? — Luminance.
46. What is measured in the SI derived unit called Tesla? — Magnetic Flux Density.
47. The unit for measuring the Wavelength of light is? — Angstrom.
48. What is measured in the unit called Gauss? — Magnetic Induction.
49. The pair represents units of energy? — Calorie and Joule.
50. Which base unit discovered in 1820 represents one Coulomb of electric charge per second? — Ampere.
51. The unit of the physical quantity Entropy is? — Joule per Kelvin.
52. The unit for measuring Sound Intensity is? — Decibel.
53. The basic SI unit of Amount of Substance is shown by? — Mole.
54. Parsec is a unit of measurement of? — Length (Distance).
55. 'Angstrom' is a unit of measurement of? — Length (Wavelength).
56. The S.I. unit of Electric Charge is? — Coulomb.
57. 1 Pascal is equal to? — 1 Newton per square meter.
58. Diopter is the unit of? — Power of a lens.
59. The unit of the physical quantity Electrical Conductance is? — Siemens.
60. The unit of the physical quantity "Jerk" is? — Meter per second cubed (m/s3).
61. The unit of the physical quantity "Magnetic Flux" is? — Weber.
62. The unit of the physical quantity "Magnetic Field Strength" is? — Ampere per meter.
63. The unit of the physical quantity 'Inductance' is? — Henry.
64. The unit of the physical quantity 'Young's Modulus' is? — Pascal.
65. The unit of the physical quantity "Heat Capacity" is? — Joule per Kelvin.
66. The unit of the physical quantity "Capacitance" is? — Farad.
67. The unit of the physical quantity (Radioactive) Activity is? — Becquerel.
68. The unit of the physical quantity Momentum is? — Newton-second.
69. The unit of the physical quantity Illuminance (Jyoti) is? — Lux.
70. The unit of Relative Density is? — It has no unit.
71. For which physical quantity is the unit Pascal NOT used? — Moment of Inertia.
72. The SI (International System of Units) unit of Length is? — Meter.

73. The SI unit of Mass is? — Kilogram.
74. The basic SI unit of Thermodynamic Temperature is? — Kelvin.
75. The formula for Pressure in physics is? — Force / Area.
76. The unit of which quantity is named after Blaise Pascal? — Pressure.
77. 'Torr' is the unit of? — Pressure.
78. The unit that measures the same quantity as the SI unit 'Pascal'? — Torr.
79. For which quantity does the unit of time appear twice in the denominator? — Acceleration (m/s2).
80. The SI unit of Quantity of Heat is? — Joule.
81. What is the relation between atm and bar? — 1 atm = 1.013 bar.
82. 1 Pound is equal to how many Ounces? — 16 Ounces.
83. Mohs scale is used for? — Measuring the hardness of minerals.
84. One Nanometer is equivalent to how many meters? — 10−9 meters.
85. What is used by meteorologists to measure solar radiation on a hemispherical area of incidence on a flat surface? — Pyranometer.
86. What kind of quantity is Electric Field? — A vector quantity.
87. When are two vectors equal? — When both their magnitude and direction are the same.
88. The quantity or phenomenon which has two independent properties - magnitude and direction, is called? — Vector.
89. What kind of quantity is Young's Modulus? — Scalar quantity.
90. What kind of quantity is Torque? — It is a vector quantity.
91. The unit of Pressure is? — Pascal.
92. The unit of Power of a Lens is? — Diopter.
93. The unit of Heat is? — Joule.
94. The unit of activity of a Radioactive substance is? — Becquerel.
95. The unit of Force is? — Newton.
96. The unit of Temperature is? — Kelvin.
97. The unit of Energy is? — Joule.
98. What does an Anemometer measure? — Wind speed.
99. What does an Ammeter measure? — Electric current.
100. What does a Tachometer measure? — Speed of rotation.

101. What does a Pyrometer measure? — High temperature.
102. High velocity is measured in? — Mach.
103. Wavelength is measured in? — Angstrom.
104. Pressure is measured in? — Pascal.
105. Energy is measured in? — Joule.
106. The unit of Acceleration is? — Meter/second^2.
107. The unit of Force is? — Newton.
108. The unit of Work done is? — Joule.
109. The unit of Impulse is? — Newton-second.
110. What is measured in Cusec? — Rate of fluid flow.
111. Byte is related to? — Computer.
112. Richter scale measures? — Intensity of earthquake.
113. Bar is the unit of? — Pressure.
114. The unit of Sound Intensity is? — Decibel.
115. The unit of Frequency is? — Hertz.
116. Watt is the unit of? — Power.
117. Knot is the unit of? — Speed of a marine ship.
118. Nautical Mile is the unit of? — Navigation distance.
119. Calorie is the unit of? — Heat.
120. Which instrument is used to measure the density of liquids? — Pycnometer.
121. Used to measure Humidity in the atmosphere? — Hygrometer.
122. Used to measure the density of milk? — Lactometer.
123. Radioactivity is measured by? — Geiger-Muller Counter.
124. Which instrument is used to measure the depth of the ocean? — Fathometer.
125. Terminal Doppler Weather Radar (T.D.W.R.) is used for? — Landing aircraft in bad weather.
126. The instrument used to measure absorbed or evolved heat? — Calorimeter.
127. Which instrument is used to measure Rainfall? — Rain Gauge.
128. Used to measure the force and velocity of wind? — Anemometer.
129. The formula for calculating BMI is? — Weight (in kg) / (Height)2 (in meters2).
130. Can be used for lie detection during investigation? — Narco Test, Brain Mapping Test, Polygraph Test.

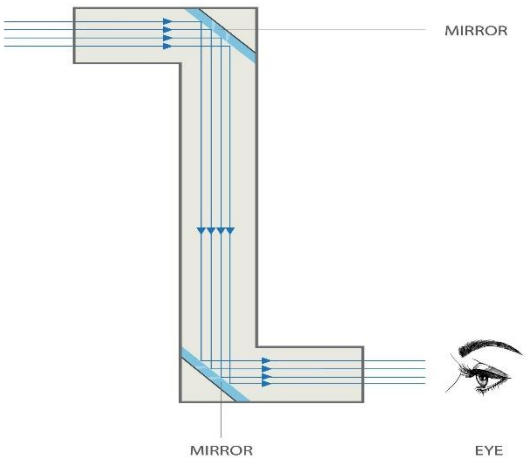
131. The instrument used to measure distance on a map? — Opisometer.
132. The instrument measuring Atmospheric Pressure is? — Barometer.
133. Ultramicro time intervals (less than a nanosecond) are accurately measured by? — Atomic Clock.
134. What is a Pyrometer called? — Radiation Thermometer.
135. Pyrheliometer is used to measure? — Solar Radiation.
136. Can be used to measure temperatures above 1500°C? — Pyrometer.
137. Used to measure extremely high temperatures? — Pyrometer.
138. The thermometer suitable for measuring 2000°C is? — Total Radiation Pyrometer.
139. 'SONAR' is mostly used by? — Navigators.
140. The intensity of an earthquake is measured by? — Seismograph.
141. Used in wristwatches, display devices, and pocket calculators? — Liquid Crystals.
142. The name of the instrument for measuring blood pressure is? — Sphygmomanometer.
143. On what principle does the Stethoscope used by doctors work? — Reflection/Superposition of sound waves.
144. Auxanometer uses? — To measure the growth rate of plants.
145. Mercury is generally used in thermometers because it has the characteristic of? — High Conductivity.
146. Used to measure the height of an object? — A Sextant.
147. The instrument used to measure the intensity of light is called? — Luxmeter.

- What kind of quantity is Young's modulus? — Scalar quantity.
- What kind of quantity is Torque? — Vector quantity.
- Why is Mercury used in thermometers? — Due to High Conductivity.
- Instrument used to measure the height of an object? — Sextant.
- Instrument measuring Light Intensity? — Luxmeter.

1. In what is the depth of the sea usually measured? Answer: Fathom.
2. What does a Polygraph machine measure in lie detection? Answer: Brain activity and heart rate.
3. Sound increases in a doctor's stethoscope, the reason is? Answer: Reflection of Sound.

4. Which waves/rays are used in Sonography? Answer: Ultrasonic waves.
5. 'Femto' means, the power of ten? Answer: -15 (10−15).
6. What does a Wind Vane instrument determine? Answer: Wind direction.
7. Used to detect acoustic signals underwater? Answer: Hydrophone.
8. What can be measured by the Reaumur Scale? Answer: Temperature.
9. Which instrument is used to measure electric current, whether it is Direct Current or Alternating Current? Answer: Ammeter.
10. An instrument used to measure the electric potential difference (Voltage) between two points in an electric circuit? Answer: Voltmeter.
11. The medical instrument used to listen to sounds generated inside the body, mainly in the heart or lungs? Answer: Stethoscope.
12. Which scientific instrument is used to measure the altitude of an aircraft above a certain level? Answer: Altimeter.
13. The name of the instrument used to measure the speed and rate of flow of fluid flowing in a pipe is? Answer: Venturimeter.
14. The following instrument is used to identify and record earthquakes? Answer: Seismograph.
15. A meteorological instrument containing a funnel-shaped collector, used to measure rainfall per unit area? Answer: Standard Rain Gauge.
16. Which instrument is used in submarines to see objects above sea level? Answer: Periscope.

SIMPLE PERISCOPE



17. The meter that measures the speed of a vehicle is called? Answer: Speedometer.
18. Sextant instrument is used in? Answer: Navigation.
19. An instrument that can be used to test whether an object is carrying an electric charge or not? Answer: Electroscope.
20. Relative humidity is measured by? Answer: With the help of Hygrometer.
21. Which instrument is used to measure the intensity of light produced by an unknown source in reference to a standard source? Answer: Photometer.
22. The instrument for measuring rain is called? Answer: Hyetometer.
23. Eudiometer measures? Answer: Volume of gas.
24. What is measured by the Beaufort Scale? Answer: Wind speed.
25. Sphygmomanometer measures the blood pressure of? Answer: Arteries.
26. Which instrument is used for the photography of the Sun? Answer: Spectroheliograph.
27. The sensory receptor related to detecting blood pressure is? Answer: Mechanoreceptor.
28. The instrument for measuring low temperature is called? Answer: Cryometer.
29. The instrument for measuring Work done is called? Answer: Ergometer.
30. The instrument used to detect and measure all types of radiation (Alpha, Beta, and Gamma) is? Answer: Geiger Counter.
31. The instrument to measure the blueness of the sky or sea is called? Answer: Cyanometer.
32. The instrument to measure Time is called? Answer: Chronometer.
33. The instrument to measure the intensity of light is called? Answer: Lucimeter.
34. The tool used to measure specific gravity is called? Answer: Aerometer.
35. We will measure the specific gravity of milk using? Answer: Using a Hydrometer (Lactometer).
36. The following instrument is used to detect current in an electric circuit? Answer: Galvanometer.
37. The following instrument is used to measure electrical resistance? Answer: Ohmmeter.
38. What does an Ammeter measure? Answer: Electric current.
39. What does a Hygrometer measure? Answer: Relative humidity.
40. What does a Barometer measure? Answer: Pressure.
41. What does a Spring Balance measure? Answer: Weight.
42. Stethoscope is used for? Answer: To listen to heart sounds.

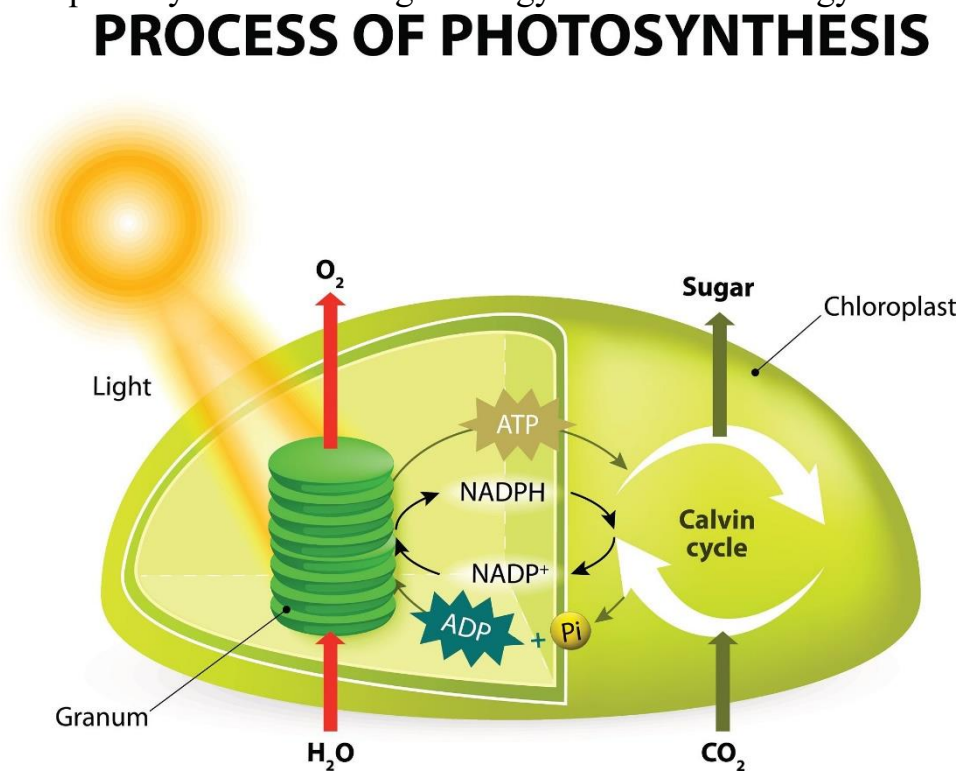
43. Sphygmomanometer is used for? Answer: To measure Blood Pressure (BP).
44. Karatmeter is used for? Answer: To detect the purity of gold.
45. Lux meter is used for? Answer: To measure light intensity.
46. What does an Anemometer measure? Answer: Wind velocity.
47. What does a Seismograph measure? Answer: Earthquake.
48. What does a Barograph measure? Answer: Atmospheric pressure.
49. What does a Hygrometer measure? Answer: Atmospheric humidity.
50. Rolling a stone up a sloping road is easier than lifting it vertically upwards, because? Answer: The work done in rolling the stone is less than lifting it.
51. What type of energy does the flowing water of a river convey? Answer: Kinetic.
52. Passengers lean forward when a moving bus suddenly applies brakes, this follows? Answer: Newton's First Law.
53. Based on what does a Jet Engine work? Answer: Conservation of Linear Momentum.
54. Sliding friction is? Answer: Slightly less than Limiting friction.
55. The time period of a simple pendulum depends on? Answer: On the length of the pendulum.
56. Motion that repeats itself after some time is called? Answer: Periodic Motion.
57. Where is gravity not present? Answer: Near an artificial satellite (Microgravity).
58. A passenger sitting in a train running at a constant speed throws a ball vertically upwards. That ball will fall? Answer: In the passenger's hands.
59. The value of 'g' (gravitational acceleration) is minimum at? Answer: At the Equator.
60. An unused book remained on the shelf for two years until it was moved. This is an example of? Answer: Newton's First Law of Motion.
61. "I throw a ball to hit the wall and it comes back to me." This is an example of? Answer: Newton's Third Law of Motion.
62. The velocity with which a satellite escapes the Earth's gravitational force is called? Answer: Escape Velocity.
63. Sound waves generated in a Sitar are? Answer: Stationary and Transverse type.
64. "For every action, there is an equal and opposite reaction", this law was given by? Answer: Newton.
65. The apparent weight of a body in a lift is more than its actual weight if the lift? Answer: Is accelerating upwards.

66. A weightless rubber balloon is filled with 200 grams of water. Its weight in water will be? Answer: Zero.
67. When a body is transferred from the Moon to Earth then? Answer: Its weight increases on Earth and Mass remains unchanged.
68. When an object is taken from Earth to the Moon then? Answer: Its weight decreases.
69. What will remain unchanged on changing the Quantity/Mass of an object? Answer: Density.
70. If an unsaturated sugar solution is prepared by mixing 100 grams of sugar in half a liter of water, then the physical quantity that will not change is? Answer: Volume.
71. Pendulum clocks become slow in summer because? Answer: The length of the pendulum increases, due to which the time taken in a unit oscillation increases.
72. When we pull a bucket of water up from a well, we feel that the bucket? Answer: Has become heavier above the water surface.
73. The Leaning Tower of 'Pisa' does not fall, because? Answer: The vertical line passing through the center of gravity remains inside the base.
74. If a bomb is dropped from an airplane moving at a uniform speed, then the position of the airplane by the time the bomb reaches the bottom will be? Answer: Exactly above the bomb.
75. The highest temperature inside a pressure cooker will depend on? Answer: On the area of the hole at the top and the weight placed on it.
76. Ink starts leaking from a pen while traveling by airplane because? Answer: Due to a decrease in atmospheric pressure.
77. Rocket works on? Answer: Principle of Conservation of Momentum.
78. Jet engine works on? Answer: Principle of Conservation of Linear Momentum.
79. If Earth's gravitational force suddenly disappears, the result will be? Answer: The weight of the object will become zero but the mass will remain the same.
80. If the distance between Earth and Sun were twice what it is, the gravitational force exerted by the Sun on Earth would be? Answer: One-fourth of what it is now.

- Where is the weight of an object maximum? — At the poles.
- An iron ball whose weight on Earth is 10 kg, its weight in space will be? — Zero.
- The weight of the human body is maximum at? — The poles.
- Sudden fall of atmospheric pressure indicates? — Storm.

- A person is standing in a boat in still water. If he moves towards the shore, the boat will? — Move away from the shore.
- A piece of ice is floating in a beaker filled with water. When the whole ice melts, then? — The water surface remains unchanged.
- A person is sitting in a moving train facing the engine. He tosses a coin which falls behind him. The train is moving? — With forward acceleration.
- Responsible for the flow of water in rivers is? — Force of friction (Viscosity).
- Density of water is maximum at? — 4°C.
- Maximum density of water is at? — 277 Kelvin.
- As soon as a train starts moving, the head of the passenger sitting in it leans backwards. The reason for this is? — Inertia of Rest.
- The form of energy in Wind Power that is converted into electrical energy is? — Kinetic energy.
- An iceberg floating in the sea has how much part above the sea surface? — 1/9 part.
- Astronauts cannot stand straight in a vacuum because? — There is no gravity.
- When the velocity of a moving object is doubled, its kinetic energy will be? — Four times.
- If an apple is released from a spaceship orbiting the earth, it will? — Move at the same speed along with the spaceship.
- Used in wristwatches, display devices, and pocket calculators? — Liquid crystals.
- When a person orbits the Earth in a satellite, his? — Mass remains unchangeable but weight becomes zero.
- A satellite keeps orbiting the Earth in its orbit. It does so? — Due to Centripetal force.
- An object is in static equilibrium when it? — Keeps moving with a uniform velocity.
- Galvanometer can be converted into Voltmeter by connecting with what? — High resistance in series.
- Which property of a substance is defined as the minimum amount of work required to remove a free electron from the surface of that substance? — Work Function.
- The mechanical transfer of energy to a system by an external force applied to it, or by a system, is called? — Work.
- The rate of work done is? — Power.

- Energy associated with falling coconuts, speeding cars, rolling stones, and flying planes is? — Kinetic Energy.
- Mechanical energy is the sum of Potential energy and which energy? — Kinetic Energy.
- Kinetic and Potential energies of a body are components of which energy? — Mechanical.
- An example of Gravitational Potential Energy is? — Water stored behind a dam.
- Hydroelectric power stations generate electricity by using the power of water falling on turbines and rotating the shaft. By rotating the turbine shaft, potential energy of water is converted into which energy? — Kinetic.
- The following is the energy conversion that takes place in the process called photosynthesis? — Light energy to Chemical energy.



- Windmill (Turbine) converts which energy into mechanical power? — Kinetic.
- Kinetic energy depends on? — Velocity and Mass of the moving body.
- Which scientific principle is primarily responsible for hydroelectric power generation? — Conversion of Gravitational Potential Energy into Electrical Energy.
- The inertia of an object is measured by which of its properties? — Mass.

- What is high in a body having excessive mass? — Inertia.
- An object resists the change in its state of rest or motion, this natural tendency is called? — Inertia.
- Philosophiae Naturalis Principia Mathematica, published in 1687, also known as Principia, is the great work of which scientist? — Isaac Newton.
- Which of Newton's laws gives the quantitative definition of force? — Second law of motion.
- Inertia is? — The tendency to resist change in the current state.
- A stationary object remains at rest, or if it is in motion, it remains in motion in a straight line at a constant speed, unless an external force is applied to it. This law states? — Law of Inertia.
- How many laws of motion did Isaac Newton formulate? — 3.
- A cannon recoils after firing? — Newton's third law of motion.
- Newton's first law is also known as? — Law of Inertia.
- An object will remain at rest as long as it is at rest, and if in motion, it will remain in motion, unless an external force is applied to it. This is Newton's? — First Law.
- A force acts on an object which is free to move. If we know the amount of force and the mass of the object, then Newton's second law of motion enables us to determine what of the object? — Acceleration.
- Rocket propulsion technology works on which scientific principle? — Newton's laws of motion.
- Dimensions of Force are? — [MLT−2].
- If two forces act on a body from opposite directions, then the magnitude of the net force acting on it is the ___ of the magnitude of both forces? — Difference.
- Which quantity is measured using Torsion Balance? — Force.
- The range of force having an exponent of 10−16m is? — Weak Nuclear.
- The direction of Torque is? — Perpendicular to the direction of applied force.
- The S.I. unit of Torque is? — Newton-meter.
- The straight distance between the point of application of force and the axis of rotation is called? — Moment Arm.
- A large force applied on a rotating object results in a large what? — Torque.
- In science, a push or pull acting on an object is called? — Force.

- If the uncertainty in position and momentum is equal, then the uncertainty in velocity will be? — $(1/2m)\sqrt{h/\pi}$.
- The formula for distance is? — Speed × Time.
- The given equation provides a property of the motion of a body moving on a circular path. In this equation 'v' is? — Speed.
- The distance covered by an object in unit time is called? — Speed.
- An object will be said to be in what type of motion if its speed keeps changing while moving in a straight line? — Non-uniform motion.
- The path of a projectile is called its? — Trajectory.
- When a particle is thrown vertically upwards and when it reaches the highest point, its? — Acceleration is downwards.
- If a ball is thrown upwards, the following does not change? — Acceleration.
- In a projectile motion, the horizontal range achieved is the same whether the object is projected at theta and which angle? — 90 degrees minus theta ($90^\circ - \theta$).
- A cannonball is fired. The motion of this ball is an example of? — Projectile motion.
- If an object is thrown upwards, its velocity upon reaching maximum height will be? — 0 m/s.
- During the motion of a projectile fired from the earth's surface? — The horizontal component of its velocity remains constant.
- The rate of change in velocity of an object per unit time is called? — Acceleration.
- The exponent of 'second' in the SI unit of acceleration is? — -2.
- Acceleration is equal to the rate of change of what? — Velocity.
- Acceleration is? — Inversely proportional to mass, proportional to force.
- The slope of a velocity-time graph shows? — Acceleration.
- An equation of motion is? — 2s=2ut+at2 (Note: Standard is s=ut+1/2at2, this is a variation).
- This type of friction is considered a self-adjusting force? — Static friction.
- Friction depends on the smoothness of surfaces. Friction force always ___ the applied forces? — Opposes.
- We are able to walk on earth without slipping? — Due to Frictional force.
- The force that stops the vehicle on applying brakes is called? — Frictional force.

- If an object slides on a surface, the force resisting the motion between them is called? — Friction.
- If the axis of rotation passes through an object while in motion, that motion is called? — Rotational motion.
- Two types of rotational motion are? — Spin and Orbital motion.
- If an object moves in purely rotational motion, then every constituent particle of it moves in a circle, whose center lies on a line called? — Axis of rotation.
- On a rotating object, what is uniform at every point? — Angular velocity.
- When net torque is zero, then what will become constant? — Angular momentum.

Mechanics, Gravitation & Properties of Gross Matter

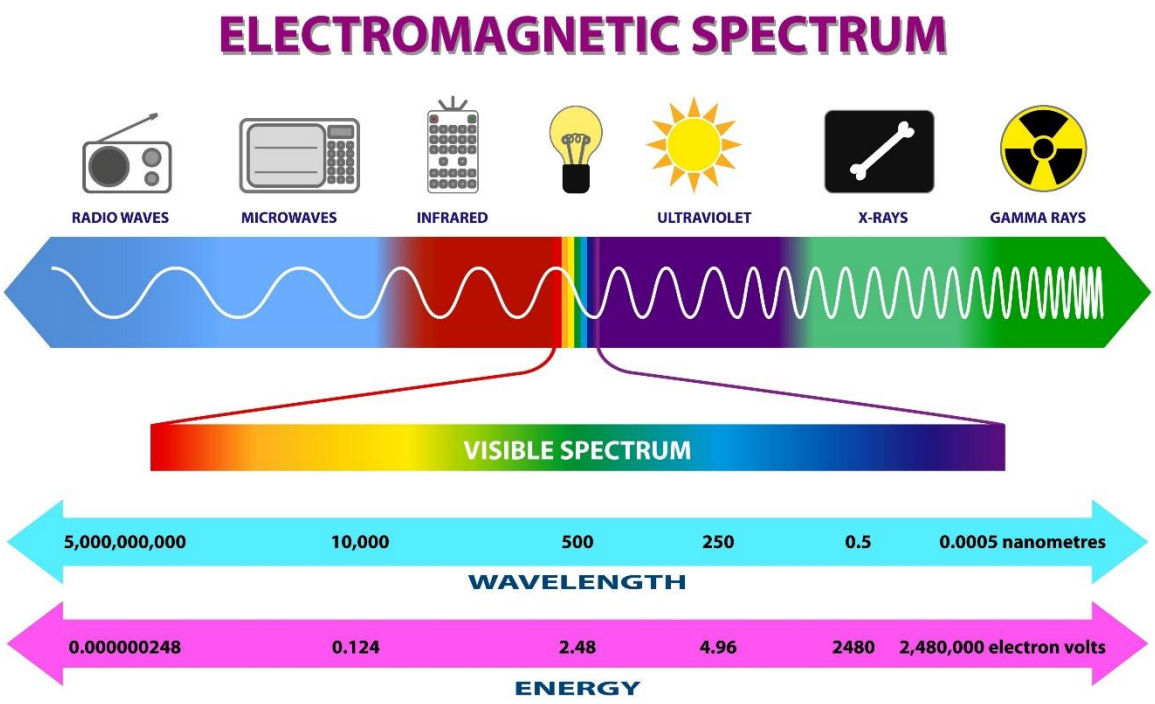
- The correct relationship between Moment of Inertia, Torque, and Angular Acceleration is? — Angular Acceleration = Torque / Moment of Inertia.
- The proper relationship between a particle of mass (m) performing simple harmonic motion and time period (T) is? — $T=2\pi m/k$.
- If an object is moving in a circular path, what will be its average speed if it completes one revolution in one second? — Zero (Note: This likely refers to average *velocity* as displacement is zero. Speed would be circumference/time).
- Work done by Centrifugal force is? — Zero work.
- The angle between centripetal acceleration and tangential acceleration is? — 90°.
- The potential energy (U) of a particle performing simple harmonic motion is? — U=0.5kx2.
- The time period of a simple pendulum is (where l is length of pendulum and g is gravitational acceleration)? — $2\pi l/g$.
- The characteristic which is not true for uniform circular motion? — Distance is always equal to displacement.
- The time taken by a pendulum to complete one oscillation is called? — Time Period.
- The motion of a simple pendulum is an example of? — Periodic motion.
- The time period of a pendulum on the Moon compared to its time period on Earth will? — Increase.

- In a simple pendulum, a small metal ball or a piece of stone is suspended from a rigid stand by a thread. The metal ball/stone piece is called the ___ of the pendulum? — Bob.
- 'Action-at-a-distance' force is? — Gravitation.
- The name of the force by which Earth pulls all objects towards itself is? — Gravitation.
- The law of gravitation applies to? — Any pair of two objects.
- According to Newton's law of gravitation, the force between two objects is? — Proportional to the product of their masses.
- Who first determined the value of G (Gravitational Constant)? — Lord Cavendish.
- In the context of gravitation, G is called? — Gravitational Constant.
- If the distance between two objects increases three times, the gravitational force of attraction between them? — Becomes one-ninth of the original.
- An iron ball and a wooden ball of equal radius are released from height H in a vacuum, the time taken by both of them to reach the ground is? — Exactly equal.
- If the speed of rotation of the Earth increases, the weight of the body? — Will decrease.
- g means acceleration due to Earth's gravitational power. The value of g is? — 9.8m/s2.
- The value of acceleration due to gravity (g) at a distance of 2R from the surface of the earth is (where R is the radius of the earth)? — g/9.
- Why is weightlessness felt when orbiting the Earth in a spacecraft? — Due to zero gravity (Free fall).
- If the radius of the Earth decreases and its mass remains constant, the value of "acceleration due to gravity" will be? — More.
- In an elevator going down with uniform velocity, the apparent weight of a person is? — Equal to the weight when the person is stationary.
- Acceleration due to gravity on a planet decreases? — With increase in altitude from the surface of the planet.
- The value of acceleration due to gravity at the center of the Earth is? — 0 (Zero).
- In which position will the weight of an object be maximum? — At the poles of the Earth.
- The weight of a body at the center of the Earth is? — Zero.

- If an object is dropped from a height and there is no air resistance? — Its speed will increase.
- The weight of a person located at a height of 2R from the center of the Earth (where R is the radius of the Earth)? — Becomes one-fourth.
- When an object falls freely from a certain height, it reaches the ground in 1 second, its velocity on colliding with the ground will be? — 9.8 m/s.
- The weight (W) of a body can be given by this formula? — W=mg.
- The orbital period of a planet in orbit of radius R is T. The orbital period in orbit of radius 4R will be? — 8 T.
- Once a satellite is placed in orbit, the only force controlling its motion is? — Gravitation.
- Particle with zero mass is? — Neutrino.
- Particle with fractional charge is? — Quark.
- Particle with fractional spin is? — Positron.
- Particle with integral spin is? — Boson.
- The metal which is most ductile? — Gold.
- Who discovered the principle of water displacement? — Archimedes.
- "A small iron ball sinks in water while a ship floats on water." This principle is based on? — The principle of scientist Archimedes.
- Who established the wave nature of particles? — De-Broglie.
- The unit of sound intensity is? — Decibel.
- The reason for echo generation is the ___ of sound? — Reflection.
- When the temperature of a substance increases, the effect on the value of average kinetic energy of its molecules is? — Increases.
- When a soap bubble is charged, what happens? — Radius increases.
- The value of weight changes at different places while mass remains? — Constant.
- It is a weak force unless large mass is involved? — Gravitational force.
- It is a semiconductor? — Silicon.
- The pressure inside a soap bubble? — Is greater than atmospheric pressure.
- Oil spreads on the surface of water because? — Surface tension of oil is less than water.
- Iron needle keeps floating on the surface of mercury? — Due to Surface Tension.

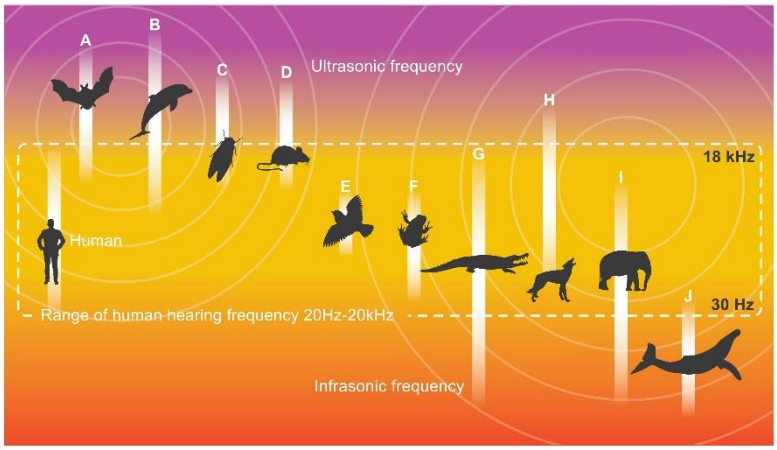
- A needle sinks in water while a ship made of iron floats on water because? — The relative density of the needle is greater than the relative density of the water displaced by it.
- When a ship enters the sea from a river? — It rises a little.
- At room temperature Silicon is a? — Semiconductor.
- Mass of an object? — Remains the same everywhere.
- Momentum of a body is defined as the product of its? — Mass and Velocity.
- What is the property of iron due to which a piece of hot iron can be beaten into the form of a knife? — Malleability.
- Spring balance works on which principle? — Hooke's Law.
- The strain produced in an object is directly proportional to the stress applied to it, this is called? — Hooke's Law.
- The principle related to the absorption of water by soil is? — Capillary action.
- A drop of water spreads on a glass plate to form a thin layer while a drop of mercury remains spherical because? — The cohesion of mercury is greater than its adhesion with glass.
- The viscosity of an ideal fluid is? — Zero.
- The reason for oil rising in the wick of a lamp is? — Capillary action.
- AV = Constant, where A = Area of cross-section and V = Velocity of fluid. This equation is called? — Equation of Continuity.
- In a streamline flow, what remains uniform at every point in the fluid? — Velocity.
- The loss in weight of a body immersed (partially or wholly) in a fluid is equal to the weight of the fluid displaced. Which of the following scientists was the first to establish this phenomenon? — Archimedes.
- Lactometers, hydrometers, ships, and submarines are designed on the basis of? — Archimedes' principle.
- For this reason, air bubbles rise in liquid? — Viscosity and Buoyancy.
- When ice floats on water, how much of it remains outside the water? — 0.1 (or 1/10th).
- If floating ice melts in a vessel, the water level in that vessel? — There is no change.
- The upward force acting on a floating object is called? — Buoyancy.

- If there is no gravity, then the following will not apply to the fluid? — Upward pressure (Buoyancy).
- Lactometer (used to determine purity of milk sample) and Hydrometer (used to determine density of liquids) are based on which principle? — Archimedes.
- The boiling point of water in mountainous areas will be? — Less than what it is at sea level.
- Why does the boiling point of water decrease at altitude? — Due to low atmospheric pressure.
- Ink pens leak in airplanes flying at high altitudes? — Due to low atmospheric pressure outside the pen.
- According to whom, pressure is equal to the force acting per unit area on which it acts? — Pascal's Principle.
- What happens when a fresh egg is put in salt water? — It floats.
- Density is maximum of? — Water (relative to ice/steam).
- The density of water is maximum at? — 4 degrees Celsius.
- When water freezes, its density? — Decreases.
- The density of the material of a substance is obtained by dividing its mass by what? — Volume.
- If some gas is removed from a closed container and the pressure is reduced, the mean free path of the gas molecules will? — Increase.
- The molecular mass of a gas is? — Twice its vapor pressure (Note: Twice its Vapor Density).
- Ultrasonic waves produced in air by a vibrating quartz crystal are an example of? — Longitudinal wave motion.
- Which waves are used by bats to locate their prey? — Ultrasonic waves (Ultrasound waves).
- Which waves are used by ships to find the depth of the ocean floor? — Ultrasonic waves.
- Which waves are used to diagnose defects in the internal organs of the human body? — Ultrasonic waves.
- In which medium can sound not travel? — Vacuum.
- The electromagnetic radiation which has the longest wavelength? — Radio waves.



- A sonorous substance is? — Iron.
- The distance between any two nearest particles of the medium vibrating in the same phase is? — Wavelength.
- Which color (among given options) has the longest wavelength? — Orange color.
- Velocity of radio wave is? — 3×108 m/second.
- FM broadcast band is between which range? — VHF range.
- The speed of sound is fastest in? — Iron.
- Sound greater than which measurement is named "Noise"? — 80 Decibels.
- The unit of measuring Supersonic Speed is? — Mach.
- Sound of which frequency will be audible to humans? — 100 Hz.

Animal hearing frequency range



A.	Bat	2kHz - 120kHz	F.	Frog & Toad	50Hz - 4kHz
B.	Dolphin	75Hz - 150kHz	G.	Crocodile	16Hz - 18kHz
C.	Insect	10kHz - 80kHz	H.	Dog	64Hz - 44kHz
D.	Rat	900Hz - 79kHz	I.	Elephant	17Hz - 10.5kHz
E.	Bird	1kHz - 4kHz	J.	Blue whale	14Hz - 36Hz

- The characteristic of a sound wave that determines its pitch? — Frequency.
- To hear an echo, the minimum distance between the source and the reflecting surface should be? — Approximately 17 meters.
- The decibel level of voice in the conversation of two persons will be? — About 30 dB.
- The minimum distance to hear an echo from a wall is? — 17.2m.
- Sound cannot travel through which medium? — Vacuum.
- Sound measuring unit is called? — Decibel.
- The speed of sound is maximum in? — In Ice (Based on given options in original context, solids conduct sound fastest).
- The human ear is most sensitive? — At 1000 Hertz.
- Persistence of hearing is? — 1/10 second.
- Persistence of vision is? — 1/16 second.
- Supersonic planes fly? — With speed greater than the speed of sound.
- An important characteristic of sound 'Pitch' depends on? — Frequency.

Sound, Wave Motion & Heat

- Clarifies the small divisible sound units of sounds, each of which has an identity — Phoneme
- The velocity of sound is maximum in — Steel
- The waves used for telecommunication are — Microwaves
- The technology used to transmit audio signals in television broadcasts is — Frequency Modulation
- Used by a remote control unit to operate a TV set — Infrared and Radio waves
- Sound is called a harmful sound pollutant if it is — Above 80 dB
- Liquid remains hot or cold for a long time in a thermos flask because there is no loss or gain of heat by — Conduction, Convection, and Radiation (by all three methods)
- Can travel in both solid and gas mediums — Sound waves
- Sound waves cannot travel in — Vacuum
- Ultrasonic are those sounds whose frequency is — More than 20,000 Hertz
- In SONAR we use — Ultrasonic waves
- The sound of the whistle of an approaching train increases while it decreases for a train moving away. This phenomenon is an example of — Doppler Effect
- The velocity of sound is maximum in — Metal
- To hear a clear echo, the minimum distance between the reflecting surface and the listener should be — 16.5 meters
- Sound travels — Like longitudinal waves
- Bats can fly in dark nights and can also hunt. The reason for this is — They emit ultrasonic waves and are guided by them
- They can destroy insects, they can clean clothes by removing dust from them, they can be used to treat diseases and they can control automatic doors; all these are properties of — Ultrasonic waves
- Used in doing CT Scan — X-rays
- Hearing sound for some time even after closing the sound source in a hall is called — Reverberation
- Mainly used to detect the presence and position of objects by radio waves — RADAR
- The wave that can destroy insects — Ultrasonic wave
- The wave that can clean clothes by removing dust — Ultrasonic wave
- The wave that can treat diseases and control automatic doors — Ultrasonic wave

- The length of waves is shortest in — γ-rays (Gamma rays)
- Waves transmit from one place to another — Energy
- The walls and ceiling of the auditorium are covered with fibrous material like glass fiber — To absorb sound and prevent echo
- An astronaut cannot hear his co-traveler on the surface of the moon because — There is no medium for sound to propagate
- The combined effect of mass and velocity is taken into account by a physical quantity called — Momentum
- Impulse is equivalent to — Change in momentum

Electromagnetic Waves & Optics

- The rate of change of momentum is — Force
- The correct order of radiations in descending order of frequencies is — X-rays > Infrared > Microwaves > Radio waves
- Which type of radiation refers to a part of the electromagnetic spectrum with frequencies of about 300 GHz to 3kHz, or equivalent wavelengths of about 1mm to 100 km — Radio waves
- Is not an electromagnetic wave — Cathode-ray
- TV remote control works on the principle of — Infrared waves
- Ionizing radiation has enough energy to affect atoms in a living cell, thereby damaging their genetic material — Radiation
- The wavelength of infrared waves is smaller than — Radio waves
- Who proposed the wave theory of light — Christiaan Huygens
- Which electromagnetic waves are detected using a thermopile — Infrared waves
- The correct order of electromagnetic radiations arranged in increasing order of frequency is — Long radio waves < Microwaves < Visible light < X-rays
- Which electromagnetic waves are produced during the radioactive decay of nuclei — Gamma rays
- In what form does the Earth's surface radiate heat into space — Infrared radiation
- Magnetron valve is used in the production of — Microwaves
- Which electromagnetic waves can be detected using a point contact diode — Microwaves

- The range of wavelength is from 700 nm to 400 nm — Visible light
- The loudness of a wave is proportional to the — Square of the amplitude
- The rate of transfer of energy per unit area perpendicular to the direction of travel of a wave is called — Intensity
- The phenomenon that established the transverse nature of light is — Polarization
- Which waves cannot be polarized — Ultrasonic
- An example of longitudinal waves (in gaseous medium) is — Sound waves
- According to the principle of superposition, the reference to the superposing of two or more coherent waves on top of each other to produce regions of maxima and minima in space is — Interference
- The proper relationship between the frequency (f) and time period (T) of a wave is — $f \times T=1$
- Geiger tube is used for the detection of which electromagnetic waves — Gamma rays
- Which are also known as heat waves — Infrared waves
- The waves used in radar systems are — Microwaves
- In which year was the Doppler effect discovered by the Austrian scientist Christian Doppler, describing the change in frequency of any type of sound or light wave produced by a moving source with respect to an observer — 1842
- The loudness of sound is proportional to the _____ of the amplitude of vibration producing the sound — Square

Sound, Heat & Thermodynamics

- Sound, which is a mechanical wave, cannot travel in which medium — Vacuum
- In which of the following media can sound not travel — Vacuum
- On what principle does the stethoscope work — Reflection of sound
- The speed of sound waves is maximum in which medium at a specific temperature — Solid
- In what does sound travel fastest — Steel
- Ultrasonic sounds are those sounds which have frequencies — Greater than 20 kHz
- The shrill voice of women is due to — High frequency
- The audible range of sound for an average adult person is — 20 Hz - 20000 Hz
- The sound of thunder is heard long after its flash. What does this indicate — The speed of sound is much less than the speed of light

- The velocity of sound in air is — 343 m/sec
- When the speed of an object exceeds the speed of sound, it is said to be traveling at — Supersonic speed
- Which characteristic of sound is measured in Hertz (Hz) — Frequency of sound
- The quality of musical sound produced by a stringed instrument depends on — Waveform of the sound
- Which gas is filled in incandescent bulbs to increase the life of the filament — Nitrogen
- To hear a clear echo, the time interval between the original sound and the reflected sound must be at least — 0.1 second
- The permissible noise level for a residential area is — 55 dB
- The permissible noise level for a silent zone is — 50 dB
- The permissible noise level for an industrial area is — 70 dB
- The permissible noise level for a commercial area is — 65 dB
- The temperature at which the density of water is maximum — 4°C
- The color of stars depends on — Their temperature
- The heat transfer method by which the Earth receives energy from the Sun — Radiation
- Cooking in a pressure cooker is faster compared to other traditional methods because — High pressure and higher boiling point inside the cooker enable faster cooking
- At what temperature is the density of pure water maximum — 4°C
- When water freezes, its density — Decreases
- Mixing salt with ice causes the freezing point to — Decrease
- The lowest temperature at which a substance catches fire is called — Ignition temperature
- The metal which is a good conductor of heat but bad for electricity — Mica
- Two glass tumblers stick to each other (one inside the other). They can be separated by — Putting the outer glass in hot water

Questions & Answers (Heat)

- Question: What is the latent heat of ice? Answer: 80 Cal/gm
- Question: What is the boiling point of water in the Fahrenheit scale? Answer: 212°F

- Question: What is the temperature of a healthy human? Answer: 37° Centigrade
- Question: The temperature at which water vapor in the air begins to condense is called? Answer: Dew point
- Question: The process by which water vapor in the air turns into water droplets is called? Answer: Condensation
- Question: What happens as a result of evaporation? Answer: Cooling
- Question: The process of vertical heating of the atmosphere is called? Answer: Convection
- Question: The process of heat transfer in solids is known as? Answer: Conduction
- Question: What is the average normal human body temperature measured by a clinical thermometer? Answer: 37°C
- Question: What is the boiling point of water in Fahrenheit? Answer: 212°F
- Question: The rate of evaporation increases with the decrease of? Answer: Humidity
- Question: What is the value of electric current in the circuit during a Short Circuit? Answer: Increases heavily
- Question: In which heat engine is the carburetor used? Answer: Petrol engine
- Question: What is the descending order of energy of electromagnetic radiations? Answer: Gamma > Beta > Alpha
- Question: Why do damp biscuits become crisp when kept inside the fridge for a while? Answer: Because the humidity inside the fridge is low and hence excess moisture is absorbed
- Question: In which condition will wet clothes dry fastest? Answer: 20% humidity and 60°C temperature
- Question: What is the safe temperature for keeping food fresh in the refrigerator? Answer: 4°C
- Question: Why is it more comfortable to wear white clothes in summer? Answer: Because they reflect all the heat falling on them
- Question: Why do aquatic animals survive in winter when the water on the upper surface of the lake turns into ice? Answer: Because the density of water is maximum at 4°C, so water remains below the upper surface of ice

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- Question: If the temperature of water is lowered from 9°C to 3°C, what will be the effect on the volume of water? Answer: Volume will first decrease and then increase
- Question: Why does a bottle break when it is filled with water and allowed to freeze? Answer: Because water expands on freezing
- Question: Humans feel troubled by humidity and heat, what is the reason for this? Answer: Sweat does not evaporate due to humidity
- Question: Why does a fan give a feeling of comfort in hot weather? Answer: Because our sweat evaporates at a faster rate
- Question: When a glass rod is placed in steam, its length increases, but what is the effect on its width? Answer: Width increases
- Question: What causes more severe burns? Answer: Steam
- Question: Who is the best conductor of heat? Answer: Water (Note: Based on options typically given in exams, otherwise metals like Silver/Copper are better).
- Question: What is the effect on volume when ice melts? Answer: Volume decreases
- Question: If an iron ball is heated, the maximum percentage increase will be in? Answer: Its volume
- Question: Why are cloudy nights warmer than clear sky nights? Answer: Because clouds prevent radiation of heat from the earth and air
- Question: What is the function of a thermostat in a refrigerator? Answer: To maintain a uniform temperature
- Question: If it takes exactly one minute for tea in a cup to cool from 90°C to 80°C, how long will it take to cool from 70°C to 60°C? Answer: More than one minute (According to Newton's Law of Cooling)
- Question: If the door of a refrigerator kept in a closed room is kept open, what will be the effect on the room temperature? Answer: Temperature will increase
- Question: If we keep heating a piece of iron, it first glows light red, then white, and finally turns which color? Answer: Orange (Note: Sequence typically goes Red -> Orange -> Yellow -> White -> Blue).
- Question: In which area is an air cooler more effective? Answer: In desert areas (dry areas)

- Question: Why does fat separate from milk in a cream separator machine? Answer: Due to Centrifugal force
- Question: Which principle of Newton is proven by the flying of birds? Answer: Third law of motion
- Question: Absolute zero can be called the temperature at which? Answer: Molecular motion of a gas stops
- Question: The kinetic energy of a 'body' is proportional to? Answer: Its momentum (specifically square of momentum)
- Question: What is the function of a thermostat in a refrigerator? Answer: Maintaining a uniform temperature
- Question: When two blocks of ice are pressed together, they form one block of ice, why? Answer: Because the melting point of ice decreases with increase in pressure
- Question: Two small drops of water merge into each other to form a large drop, what happens in this process? Answer: Energy is released
- Question: Which device is used to measure temperature? Answer: Thermometer
- Question: A clinical thermometer measures temperature in which range? Answer: 35°C to 42°C
- Question: What is the Steam Point on the Fahrenheit scale? Answer: 212°F
- Question: What is the boiling point of water at sea level? Answer: 100°C
- Question: How does the temperature remain at the boiling point of a liquid? Answer: Temperature remains constant
- Question: 30° Celsius is equal to how many Kelvin? Answer: Approx 303 Kelvin (273+30)
- Question: The direction of flow of heat between two bodies depends on? Answer: Their temperature
- Question: What is the reading of temperature where both Fahrenheit and Celsius scales are the same? Answer: −40°
- Question: 'Absolute zero' is the temperature at which what happens? Answer: Molecular motion in gases starts decreasing / All molecular motion stops
- Question: 0°K is equal to? Answer: −273°C

- Question: When the temperature difference between the liquid and its surroundings is doubled, the rate of loss of heat will be? Answer: Doubled
- Question: Cryogenics is a branch of physics related to? Answer: Extremely low temperatures
- Question: The difference in temperature of two bodies is 30 degrees Centigrade, what will be this difference in degrees Fahrenheit? Answer: 54 (30×1.8)
- Question: By which medium does the Sun give heat to the Earth? Answer: By radiation
- Question: In what form is the latent heat energy stored during temperature change in a substance? Answer: Kinetic energy (Potential energy is more accurate for latent heat phase change, but kinetic is mentioned in source)
- Question: Why does the air near the surface of the earth continue to receive heat even after sunset? Answer: Terrestrial radiation
- Question: One calorie of thermal energy is equal to approximately how many Joules of mechanical energy? Answer: 4.2 Joules
- Question: What causes the Eiffel Tower to become taller during summers? Answer: Thermal expansion
- Question: What is the horizontal heat transfer within the atmosphere called? Answer: Advection
- Question: Why does food cook faster in a pressure cooker? Answer: Due to increase in boiling point of water
- Question: A wooden spoon is dipped in a cup of ice cream, how will its other end be? Answer: Will not become cold
- Question: Who is the best insulator? Answer: Glass

Thermodynamics

- Question: Why does ice covered with sawdust not melt quickly? Answer: Because sawdust is a bad conductor of heat.
- Question: Who coined the term 'Zeroth Law of Thermodynamics' in 1931? Answer: Ralph H. Fowler.
- Question: In thermodynamics, what is the type of energy that transfers from a hot substance to a cold substance? Answer: Heat.

- Question: If the temperature of two bodies is the same, in what state are they considered to be? Answer: In Thermal Equilibrium.
- Question: What is the thermodynamic process called in which the volume of a closed system remains constant? Answer: Isochoric Process.
- Question: When water freezes to form ice, what is the effect on its Entropy? Answer: Entropy decreases.
- Question: Which law of thermodynamics states that "the increase in internal energy is equal to the sum of the heat supplied and the work done by the system"? Answer: First Law of Thermodynamics.
- Question: Which scientist formulated the first two laws of thermodynamics and deduced that the absolute zero of temperature is –273.15°C? Answer: William Thomson (Lord Kelvin).
- Question: What is the Boiling Point of water on the Kelvin scale? Answer: 373 K.
- Question: What is the normal human body temperature in Kelvin? Answer: 310 K.
- Question: At what temperature is the density of water maximum? Answer: At 277 K (4∘C).
- Question: –40°F temperature is equal to what on the Kelvin scale? Answer: Equal to 233 K.

Optics (Light & Optics)

- Question: What is the center of the reflecting surface of a spherical mirror called? Answer: Pole.
- Question: What is the unit of measuring Stellar Distance? Answer: Light Year.
- Question: Why does the clear sky appear blue? Answer: Due to Scattering of light.
- Question: When a green leaf is viewed in red light, what color will it appear? Answer: Black.
- Question: When the object is placed at infinity, what is the nature of the image formed by a convex mirror? Answer: Virtual and Erect.
- Question: What is the phenomenon of deflection (bending) of light called when it enters obliquely from one medium to another? Answer: Refraction.

Light & Mirror

- Question: The lens used by people with myopia (near-sightedness) is? Answer: Concave Lens

- Question: A patient with hypermetropia (farsightedness) cannot see objects closer than 40 cm clearly. The lens required to make the eye normal will be? Answer: Convex lens of focal length 200/3 cm
- Question: The minimum height of a plane mirror required to see the full image of a person is? Answer: Half of the person's height
- Question: When a red rose is viewed through a blue filter, it appears? Answer: Black
- Question: The real image of an object is formed by a convex lens on a screen. If the upper part of the convex lens is covered with opaque paper, what will be the effect on the image? Answer: The complete image of the object will be obtained, but its intensity will decrease
- Question: Which color of light has the shortest wavelength? Answer: Violet
- Question: Which light has the maximum energy? Answer: In Blue light (or Violet)
- Question: The best colors for a sun umbrella would be? Answer: White on top and Black on the inside
- Question: Who has the highest refractive index? Answer: Diamond
- Question: The reason for the sky appearing blue is? Answer: Scattering of light
- Question: The color of light that travels fastest in a medium other than vacuum or air? Answer: Red
- Question: The mirror used to see side objects in motor vehicles is? Answer: Convex Mirror
- Question: The reason for sea water being blue is? Answer: Scattering of light by water
- Question: Why is oil immersion used in an optical microscope? Answer: To increase the resolving power of the microscope
- Question: What is formed when an opaque object comes in the path of light? Answer: Shadow
- Question: Primary colors are? Answer: Red, Green, Blue
- Question: Astigmatism is corrected by using which lens? Answer: Cylindrical lens
- Question: Used by dentists to view an enlarged image of teeth? Answer: Concave mirror
- Question: LED is the current technology for energy-rich light. What is the full form of LED? Answer: Light Emitting Diode

- Question: What is the 'Very Light Pistol' used for at night? Answer: To illuminate a place for a short time
- Question: Who propounded the wave theory of light? Answer: Huygens
- Question: When a red rose is viewed through green light, it appears? Answer: Black
- Question: Not an advantage of optical communication? Answer: High speed (Note: This is context-dependent, typically optical has high speed, but this might refer to latency in specific scenarios or a tricky option).

Scattering, Refraction and Human Eye

- Question: The reason for the propagation of light in the atmosphere is? Answer: Dust particles (Scattering)
- Question: Who reflects more sunlight? Answer: Land covered with fresh snow (High Albedo)
- Question: How to connect two identical bulbs to get more light? Answer: In Parallel
- Question: Solar energy is converted into electrical energy? Answer: By Photovoltaic cells
- Question: The formation of Mirage is an example of? Answer: Total Internal Reflection
- Question: A black body does not do? Answer: Refraction of radiation
- Question: Einstein was given the Nobel Prize for? Answer: Theory of Photoelectric effect
- Question: That whose velocity is maximum? Answer: Light
- Question: In hypermetropia (farsightedness), the image is formed? Answer: Behind the retina
- Question: Time taken for light to reach the earth from the sun is? Answer: About 8 minutes
- Question: We get sunlight in the room? Answer: By Scattering
- Question: Visible due to dispersion of sunlight? Answer: Rainbow
- Question: Appears in the middle of the rainbow? Answer: Green color
- Question: The number of colors in sunlight is? Answer: 7
- Question: Doctors use an endoscope to see the internal organs of the body. It is based on? Answer: Principle of total internal reflection of light

- Question: Air bubble inside water behaves like? Answer: Like a Concave lens (Diverging lens)
- Question: A cut diamond shines with sparkle? Answer: Due to total internal reflection
- Question: Photo-voltaic cells are? Answer: Solar-cells
- Question: The minimum distance of clear vision for a normal healthy human eye is considered to be? Answer: 25 cm
- Question: Used to see traffic coming behind the car? Answer: Convex mirror
- Question: If a point source of light is placed between two parallel plane mirrors, images will be formed? Answer: Infinite
- Question: Full form of C.F.L. is? Answer: Compact Fluorescent Lamp
- Question: The color of the sky is blue? Answer: Due to scattering of light
- Question: Objects are visible in light? Answer: Due to reflection
- Question: The mirror used by dentists to see patients' teeth is? Answer: Concave
- Question: A person with near-sightedness 'Myopia' can see? Answer: Objects located nearby clearly
- Question: Myopia is corrected? Answer: Using concave lens
- Question: The cause of Mirage is? Answer: Total internal reflection of light
- Question: The color of light is determined by its? Answer: Wavelength
- Question: The reason for the Sun and Moon appearing elliptical near the horizon is? Answer: Refraction
- Question: The image formed on the retina is? Answer: Smaller than the object and inverted
- Question: Diamond appears shiny? Answer: Due to Total Internal Reflection
- Question: In the human eye, light waves are converted into nerve impulses? Answer: On the Retina
- Question: Focusing happens in the eye? Answer: By changing the convexity of the lens
- Question: The human eye is most sensitive to which wavelength of visible light? Answer: 5500 Angstroms (Yellow-Green)
- Question: Power of sunglasses is? Answer: 0 Diopter

Properties and Applications of Light

- Question: Red light is used for danger signals because? Answer: Its scattering is the least

- Question: Color combination in an umbrella to protect from the sun is appropriate? Answer: White on top, black below
- Question: The color of light is determined by its? Answer: Wavelength
- Question: The concept of 'Expanding Universe' is based on? Answer: Doppler effect
- Question: The reason for the red color of the sun at sunset and sunrise is? Answer: Scattering of light
- Question: The image formed by an astronomical telescope is? Answer: Virtual and Magnified
- Question: The mirror used in a Searchlight is? Answer: Concave mirror
- Question: The sensitivity of the eye is maximum? Answer: Between Yellow and Green colors
- Question: The wavelength range of visible light is? Answer: Between 390-780 nm
- Question: Do not use in night vision cameras? Answer: X-ray radiation from objects (They use IR)
- Question: The color of a star indicates? Answer: Its temperature
- Question: When light goes from one medium to another, what remains unchanged? Answer: Frequency
- Question: Looking directly at the sun during a total solar eclipse causes irreversible damage to the eye. Burning of the retina is caused by which ray of the sun? Answer: Ultraviolet (UV) light
- Question: When a pencil-like object is rotated rapidly in front of a running TV screen or tubelight, multiple images of that object are seen. The reason for this is? Answer: Stroboscopic effect
- Question: 'Blind spot' and 'Yellow spot' are found in the eye? Answer: In the Retina
- Question: The image formed on the retina is? Answer: Smaller than the object and inverted
- Question: Scientists have recently created the world's smallest light bulb? Answer: With the help of Graphene
- Question: To see the full image of a person, the minimum height of a plane mirror should be? Answer: Half of the person's height
- Question: The radius of curvature of a plane mirror is? Answer: Infinite
- Question: The velocity of light is maximum? Answer: In Vacuum

- Question: The alkaloid whose very dilute solution is used to dilate the pupil of human eyes is? Answer: Atropine
- Question: Light appears to travel in a straight line because? Answer: The wavelength of light is very small
- Question: The best energy source from the point of view of keeping the environment clean is? Answer: Solar cell
- Question: A rod immersed in a vessel filled with water appears bent? Answer: Due to refraction
- Question: Immediately before sunrise or sunset, the sun appears oval? Answer: Due to the effects of Refraction
- Question: When an object is placed between two parallel plane mirrors, the number of images formed is? Answer: Infinite
- Question: Red light is used for danger signal because? Answer: Its scattering is the least
- Question: Laser is a device by which is generated? Answer: Stimulated radiation
- Question: The effect which helps in determining the nature of the universe? Answer: Doppler's effect
- Question: Which color of light of VIBGYOR has the minimum wavelength? Answer: Violet
- Question: How many colors are contained in sunlight? Answer: Seven
- Question: Apart from the wavelengths of red and green color of sunlight, which other color wavelength of sunlight is absorbed by water molecules in the oceans? Answer: Yellow
- Question: The electromagnetic spectrum describes the entire range of light which is generally divided into how many regions of wavelength and in order of increasing energy and frequency? Answer: Seven
- Question: Which color light is formed when red, blue and green lights are mixed in equal amounts? Answer: White
- Question: Light waves are? Answer: Electromagnetic waves
- Question: Light travels at the fastest speed? Answer: In Vacuum
- Question: Objects which emit light themselves like the sun are called? Answer: Luminous bodies/objects

- Question: The color of light emitted by the sun is? Answer: White
- Question: Scattering of a light beam by a medium containing microscopic suspended particles is called? Answer: Tyndall effect
- Question: The color of the sun appears red at sunrise and sunset? Answer: All colors except red get scattered
- Question: The sky appears blue because blue light is ..... more than other colors in the atmosphere? Answer: Scattered
- Question: The reason for the sky looking blue is? Answer: Due to shorter wavelength, blue color scatters easily
- Question: The splitting of white light into its component colors is called? Answer: Dispersion
- Question: The name of the phenomenon (driven by scattering of light) in which mountain peaks turn pink or orange at sunrise and sunset? Answer: Alpenglow
- Question: Raman wants to use a rear-view mirror in his vehicle. For this he should choose a mirror? Answer: Convex mirror
- Question: The following mirror is used as a headlight in vehicles? Answer: Concave mirror
- Question: Lateral inversion of light is displayed in the following type of mirror? Answer: Plane mirror
- Question: If light is reflected from two plane mirrors placed facing each other, then images will be formed? Answer: Infinite
- Question: The size of the image formed by a plane mirror is always ..... the object? Answer: Equal to
- Question: The image formed by a plane mirror is always? Answer: Erect
- Question: In an image formed by a mirror, the object on the left appears on the right and the object on the right appears on the left. This is called? Answer: Lateral Inversion
- Question: What will be the total number of images formed by two mirrors inclined at 120° to each other? Answer: 2
- Question: For use in a solar cooker, the mirror is most suitable? Answer: Concave
- Question: If the mirror used is a concave mirror, then the correct formula for magnification among these is? Answer: (Image distance) / (Object distance)

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- Question: Mirror used in car headlights? Answer: Concave
- Question: In a concave mirror, if the object is placed at the center of curvature, the image formed will be? Answer: Located at the center of curvature
- Question: In this mirror the image is virtual, erect and of point size? Answer: Convex
- Question: In optics, the refractive index of a substance is given by the formula n=c/v, where c is? Answer: Speed of light in vacuum
- Question: The phenomenon of change in direction of light when it passes from one medium to another is called? Answer: Refraction
- Question: When light waves enter from one medium to another, they deviate from their original path. This phenomenon is called? Answer: Refraction
- Question: Whose refractive index is closest to Crown Glass? Answer: Canada Balsam
- Question: The shine in bubbles produced in water is due to which process? Answer: Due to Total Internal Reflection of light
- Question: The focal length of a convex lens with +2.5 Diopter power will be? Answer: 40 cm
- Question: The lens in the upper part of a bi-focal lens is? Answer: Concave
- Question: The following lens always forms a virtual, erect and smaller image? Answer: Concave
- Question: If the power of an object is expressed in Dioptre, then that object must be a .....? Answer: Lens
- Question: The power of a lens is -2.0 D. Where 'D' stands for? Answer: Diopter
- Question: When you come from a bright room, you cannot see objects in a dimly lit room? Answer: The iris contracts making the pupil smaller allowing less light to enter the eye
- Question: Muscles can help in modifying the curvature in the eye lens? Answer: Ciliary muscles
- Question: People who wear bifocal lenses are found to have? Answer: Presbyopia
- Question: Which type of lens is commonly used to correct Presbyopia? Answer: Bifocal lens
- Question: Rajat has Hypermetropia (long-sightedness). Which type of lens will the ophthalmologist suggest to correct his vision? Answer: Convex

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- Question: The change in focal length of the eye lens is caused by the action of .....? Answer: Ciliary muscle
- Question: How far (in cm) does a person with hypermetropia have to keep reading material to read clearly? Answer: 25
- Question: When the human eye lens becomes thin, the focal length will? Answer: Increase
- Question: Made the first compound microscope? Answer: Zacharias Janssen
- Question: A periscope uses? Answer: Two plane mirrors
- Question: Instrument used for measurement of direct beam solar radiation? Answer: Pyrheliometer
- Question: The splitting of light into different colors is called? Answer: Dispersion
- Question: The longest wavelength in the colors of the rainbow is? Answer: Of Red color
- Question: In the process of formation of a rainbow after a rain shower, sunlight is ..... through small water droplets? Answer: Dispersed
- Question: The wavelength of the following color of the rainbow is the shortest? Answer: Violet

Electricity

- Question: According to Ohm's law, the true equation is? Answer: R=V/I
- Question: Example of an insulator is? Answer: Diamond
- Question: Converts mechanical energy into electrical energy? Answer: Dynamo
- Question: The most commonly used material for making the filament of an electric bulb is? Answer: Tungsten
- Question: Used to measure electromotive force? Answer: Potentiometer
- Question: ..... device is used to convert AC to DC (Alternating Current to Direct Current)? Answer: Rectifier
- Question: If net charge Q flows across any cross-section of a conductor in time (t), then the current (I) flowing through the cross-section will be? Answer: I=Q/t
- Question: The device which converts electrical energy into mechanical energy? Answer: Electric Motor
- Question: Transformer works on the principle of .....? Answer: Electromagnetic Induction

- Question: Fuse protects an electric circuit from? Answer: Overloading
- Question: ..... instrument is used to detect the presence of electricity in a circuit? Answer: Galvanometer
- Question: Device used to connect various appliances in series? Answer: Connecting wire
- Question: A substance which is a bad conductor of electricity but a good conductor of heat? Answer: Mica
- Question: A current flowing through a conductor is? Answer: Directly proportional to potential difference
- Question: The force exerted by a charged object on another charged or uncharged object is called? Answer: Electrostatic force
- Question: Chemical energy is converted into electrical energy? Answer: By Storage cell
- Question: The best conductor of electricity is? Answer: Sea water
- Question: Increase in light only when the speed of a bus increases indicates? Answer: Weak battery
- Question: Photo-voltaic cells are? Answer: Solar-cells
- Question: A ..... is a passive two-terminal electrical component that implements electrical resistance as a circuit element? Answer: Resistor
- Question: A simple device used to break an electric circuit or complete it? Answer: Switch
- Question: It is an electrical device that can be used to expand the capacity of a wall outlet, which can accommodate appliances? Answer: Power strip
- Question: The best conductor of electricity is? Answer: Copper
- Question: Elements of electric iron, heater, electric kettle etc. are made of? Answer: Nichrome
- Question: Heating a semiconductor makes its resistance? Answer: Decrease
- Question: The best conductor of electricity is? Answer: Sea water
- Question: 'Dynamo' is a device which converts? Answer: Mechanical energy into electrical energy
- Question: 'Semiconductor' is? Answer: Germanium

- Question: Electric motors often burn out when operating at low voltage because? Answer: They draw more current which is inversely proportional to the voltage
- In a three-pin electric plug, the longest pin should be connected to — The Earth terminal
- When the TV is switched on — Audio and video start simultaneously
- Transformers are used — To step up or step down AC voltage
- Used to step up or step down alternating current potential — Transformer
- Earth is used in electrical appliances — For Safety
- Newly invented high temperature superconductor is — Ceramic Oxides
- The characteristic of wire used in 'Fuse' is — High resistance / Low melting point
- Is a very good conductor of heat — Mercury
- Resistance of a semiconductor on heating — Decreases
- The best conductor of electricity is — Silver
- A device which converts electrical energy into mechanical energy is — Motor
- Silicon and Germanium are — Semiconductors
- Fans, bulbs, tubes etc. in houses are installed — In Parallel combination
- Electric current does not flow between two charged bodies connected together if they are — At the same potential
- The maximum temperature achieved by a superconductor is — 133 Kelvin
- An important part of a transistor is — Germanium
- Converting mechanical energy into electrical energy is the function of — Dynamo
- The characteristic of the wire used in the fuse is — High resistance - Low melting point
- Diode which emits light when current is passed through it is — LED
- The concept of electric field was first presented by — Michael Faraday
- The total charge of an Electric Dipole is — 0 (Zero)
- Correct formula to find the potential difference between two points — V=W/Q
- Electric field can be defined as — Force per unit charge
- The relationship between Capacitance C, Charge Q and Potential V is — Q=CV
- The energy stored in a capacitor of charge Q, voltage V and capacitance C cannot be expressed by the following relation — (1/2)[(V2)/Q]

- Which law of physics states that when the distance between two electric charges is doubled, the force between them is reduced to one-fourth of its previous value — Coulomb's Law
- In this situation current will stop flowing in the bulb — When the circuit breaks
- A property of the wire used to make electric fuse is — It melts quickly
- The rating for a fuse used in domestic electric circuits is provided on the basis of — Electric current
- Fuse wires are made of alloy — Tin and Lead
- Kirchhoff's first law states that the total current entering a junction is equal to the total current leaving that junction — Kirchhoff's First Law
- Precautions to be taken to prevent overloading of domestic electrical circuits — Fuse must be connected to the circuit
- Electrical wires are generally coated with — Polyvinyl Chloride (PVC)
- 'Kirchhoff's Laws' are related to which field of science — Electric Circuit
- The most important safety device in an electrical circuit, used to protect the circuit due to short circuiting and overloading — Fuse
- Is a passive two-terminal electrical component that produces electrical resistance as a circuit element — Resistor
- Coulomb per second is equal to — Ampere
- In a resistor, when the resistance is doubled the current — Current becomes half
- When an electric current flows through a wire, it happens — The wire gets hot
- Frequency of Direct Current is — Zero
- Not a good conductor of electricity — Diamond
- Formula showing Ohm's law — V=I×R (Note: Original text has a typo A=V*ohm, corrected to standard)
- Discovered the relationship between the current flowing in a metal wire and the potential difference between its terminals — Georg Simon Ohm
- Georg Simon Ohm, in whose honor we have the famous Ohm's Law, was from the country — Germany
- Electrical conductivity through metals is called electronic conduction and it — Decreases with increase in temperature
- We can compare electric current in an electric conductor to — Water currents in a river

- Substances which allow electric current to flow through them are — Electric conductors
- Is a good electric conductor — Graphite
- Copper wires are often used instead of iron for power transmission because — Copper can carry more electricity than iron (Lower resistance)
- Ohm-conductor is — Silver
- This law states that the ratio of the potential difference across a conductor to the current flowing through it is a constant, provided that its physical conditions like temperature etc. remain constant — Ohm's Law
- Is not a conductor of electricity — Ebonite
- An electric bulb lit by an 18 V battery draws 3 A electric current. The resistance of the bulb will be — 6Ω
- Substances which do not allow electric current to flow through them are called — Insulators
- If four resistors of 3Ω,4Ω,5Ω and 6Ω are connected in series, then the combined resistance of the four resistors will be — 18Ω
- If electrical resistance is to be reduced then the number of resistors should be connected — Parallel
- Resistance of a metal conductor — Is proportional to its length
- When two resistors R1 and R2 are connected in series, their equivalent resistance will be — R1+R2
- Affects the resistance of the wire — Resistivity, Length of wire, Area of cross-section of wire
- Is not a factor on which the resistivity of the wire depends — Density of the material
- When two resistors are connected in parallel, remains in them — Potential difference same
- As the temperature of a semiconductor increases, its resistivity — Decreases
- The following term does not represent electric power in an electric circuit — IR2
- Heat generated in a resistor wire is — Proportional to the square of the electric current flowing in the wire
- The coil of wire used in electric room heater is called — Element

- The following device is used in digital circuits so that the information stored in the memory of large computers is not lost during momentary power failure — Capacitor
- The instrument used to measure potential difference in an electric circuit is called — Voltmeter
- The instrument used to measure small electric current is — Galvanometer
- In an electrostatic precipitator, there is an electrode wire through which thousands of volts pass, to remove electrons, it produces — Corona
- The following device can detect the presence of electric current in a circuit — Galvanometer
- The following component cannot be made within an Integrated Circuit (IC) — Transformer
- Transformer is a device to convert — Low voltage AC to High voltage AC
- In alternating current, the following device is used to efficiently convert one voltage to another — Transformer
- The following gas is used in electric bulbs — Nitrogen
- Filaments of electric bulbs are generally made of — Tungsten
- In the symbol of an electric cell, the thick, short line represents — Negative terminal
- What is the initial voltage of batteries used for household gadgets like remote controls, small toys, and wireless kitchen appliances — 1.5 V
- A combination of two or more cells in which the positive terminal of one cell is connected to the negative terminal of the other cell is called — Battery
- A battery converts energy into electrical energy — Chemical
- The following device converts chemical energy into electrical energy — Battery
- A ..... is an electronic device that receives a weak signal and regenerates it — Repeater
- A sealed glass vacuum tube containing an electron beam focused on a phosphor-coated glass screen when light is emitted — Cathode Ray Tube
- Adding suitable impurities to a semiconductor is called — Doping
- The following component is used as an amplifier device — Transistor
- Diode is used for — Rectification
- Rectifiers convert — AC to DC
- What type of material is Silicon — Semiconductor

Magnetism & Electromagnetism

- An electric current always produces a field — Magnetic
- Electromagnet is prepared from — Soft iron
- Not attracted by a magnet — Aluminum
- Question: What is the effect on the strength of the magnetic field by increasing the number of turns in the coil? Answer: Decreases (Note: As per your source text).
- Question: If a magnet is cut into two pieces of equal length, how will they behave? Answer: Both pieces will act like a complete magnet.
- Question: Permanent magnets are made of which materials? Answer: Ferromagnetic materials.
- Question: Which scientist suggested that a magnet must also exert an equal and opposite force on a current-carrying conductor? Answer: Andre Marie Ampere.
- Question: Induced current is highest when the direction of motion of the coil is? Answer: Perpendicular to the magnetic field.
- Question: The radius of curvature of the path of a charged particle in a uniform magnetic field is proportional to? Answer: Momentum of the particle.
- Question: Which element is highly effective for making permanent magnets? Answer: Steel.
- Question: The direction of magnetic lines of force inside a bar magnet is? Answer: From S-pole to N-pole of the magnet.
- Question: Elements named Nickel and Cobalt are classified as what in terms of their magnetic properties? Answer: Ferromagnetic.
- Question: In Fleming's Left Hand Rule, which finger represents the electric current? Answer: Middle finger

- Question: The discovery and understanding of electromagnetic induction is based on a long series of experiments by whom? Answer: By Faraday and Henry.
- Question: When an electric current flows through a wire, it acts like a? Answer: Magnet.
- Question: What is Fleming's Right-Hand Rule? Answer: The motion of the conductor, the magnetic field, and the induced current are perpendicular to each other.

- Question: What is the phenomenon of electromagnetic induction? Answer: Inducing electric current in a coil due to relative motion between the coil and the magnet.
- Question: A rectangular coil of copper wire is rotating in a magnetic field. The direction of induced electric current in this coil changes after how many rotations? Answer: Half rotation.
- Question: The magnetic field produced by a coil having n turns is how many times the magnetic field produced by a single turn? Answer: n times.
- Question: When the direction of motion of the coil is perpendicular to the direction of the electromagnetic field, what is the value of the induced current generated? Answer: It is maximum.
- Question: Who formulated the accepted theory of electromagnetic radiation? Answer: James Clerk Maxwell.
- Question: The direction of magnetic field lines (outside) the magnet is? Answer: From North Pole to South Pole.
- Question: How does the magnetic field remain when moving from one end to the other inside a solenoid? Answer: Remains uniform.
- Question: A positively charged particle projected towards the west is deflected towards the north by a magnetic field. The direction of the magnetic field is? Answer: Upward.
- Question: Which law states that the induced EMF is directly proportional to the rate of change of magnetic flux linkage? Answer: Faraday's Law.
- Question: Fleming's 'Left Hand Rule' is related to? Answer: Effect of magnetic field on current.
- Question: Through a plane surface, the product of the normal magnetic flux density and the area of the surface is? Answer: Magnetic Flux.
- Question: What is defined as the force acting per unit current flow per length on a wire placed at right angles in a magnetic field? Answer: Magnetic Flux Density.
- Question: What is an electromagnet made of? Answer: Soft iron.
- Question: What is not a primary force but influences ocean currents? Answer: Earth's magnetic field.
- Question: Who discovered and named the science of Electromagnetism? Answer: Andre Marie Ampere.

- Question: The set of equations involving electric and magnetic fields and their sources, charge and current density, is known as? Answer: Maxwell's equations.
- Question: When do magnets lose their properties? Answer: On heating, on beating with a hammer, on dropping from a height.

Communication & Modern Physics

- Question: What are the three basic units of an electronic communication system? Answer: Transmitter, Transmission Channel, and Receiver.
- Question: What does 'Noise' refer to in electronic communication systems? Answer: Unwanted signals that interfere with the transmission and processing of message signals.
- Question: For which work was Albert Einstein awarded the Nobel Prize? Answer: For the law of Photoelectric Effect.
- Question: The scientist who proposed the Theory of Relativity? Answer: Albert Einstein.
- Question: What is used to make a nuclear bomb? Answer: Uranium.
- Question: Compton effect occurs in? Answer: X-rays.
- Question: Which is the first nuclear reactor in India? Answer: Apsara.
- Question: What is a nuclear property? Answer: Radioactivity.
- Question: What is used as a moderator in a nuclear reactor? Answer: Graphite.
- Question: The substance used as both moderator and coolant in nuclear reactors is? Answer: Heavy water.
- Question: Natural uranium in which the component of the radioactive U235 isotope is artificially increased is called? Answer: Enriched Uranium.
- Question: The reason for the renewable source of energy in stars is? Answer: Conversion of Hydrogen into Helium.
- Question: The Sun's energy is produced by? Answer: Nuclear Fusion.
- Question: What is the difference between a nuclear reactor and an atomic bomb? Answer: In a nuclear reactor, the chain reaction is controlled.
- Question: What is the basis of the theory of the atom bomb? Answer: Nuclear Fission.
- Question: The working principle of the atom bomb is nuclear fission of uranium and the working principle of the hydrogen bomb is? Answer: Nuclear fusion of Deuterium.

- Question: What is based on the principle of uncontrolled fusion reaction? Answer: Hydrogen Bomb.
- Question: The reactor which produces more fissionable material than it burns is? Answer: A Breeder Reactor.
- Question: Why has Kalpakkam been in the headlines? Answer: Because of its Nuclear Power Project.
- Question: What is used as a coolant in the 'Fast Breeder Test Reactor' at Kalpakkam? Answer: Molten Sodium.
- Question: What cannot be used as nuclear fuel? Answer: Calcium.
- Question: India's nuclear plant located in Seismic Zone IV? Answer: Narora.
- Question: India's twentieth nuclear power station is? Answer: Kaiga (Karnataka).
- Question: The number of member countries in the Nuclear Suppliers Group (NSG) as of July 2016 is? Answer: 48.
- Question: Reactors that operate at very high temperatures and usually have a Tokamak design are? Answer: Nuclear Fusion Reactors.
- Question: Photoelectric effect shows which nature of light? Answer: Particle nature.
- Question: Based on the quantum formulation given by Indian physicist Satyendra Nath Bose, in which year did Albert Einstein predict a new state of matter, the Bose-Einstein Condensate (BEC)? Answer: 1924.
- Question: What does D stand for in LED? Answer: Diode.
- Question: What should be the characteristic of the metal to be used as a cathode to produce the photoelectric effect? Answer: Should have low work function (Low reactivity/Low ionization energy).
- Question: What is emitted from every hot object? Answer: Infrared rays.
- Question: The name of American scientist Edwin Hubble is associated with which theory? Answer: Big Bang Theory.
- Question: Who laid the foundation of Quantum Theory? Answer: Max Planck.
- Question: The code given to the first nuclear test of Pokhran in 1974 was? Answer: Smiling Buddha.
- Question: During the radioactive decay of nuclei, which electromagnetic waves are produced? Answer: Gamma rays.

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- Question: What happens during Nuclear Fusion? Answer: Two light nuclei combine to form a heavy nucleus.
- Question: In a nuclear reactor, what role does the moderator play? Answer: It slows down the neutrons released in fission.
- Question: What is NOT used as a neutron-absorbing material in nuclear reactors? Answer: Iron.
- Question: Which substance is NOT used as a coolant in nuclear reactors? Answer: Graphite (It is used as a moderator).
- Question: Which metal is used as a shield in a nuclear reactor? Answer: Lead.

Discoveries & Inventions

- X-ray was invented by? — William K. Roentgen.
- The first missile developed in India is? — Prithvi.
- Who discovered heavy water? — Urey.
- Who invented Dynamite? — Alfred Nobel.
- Dynamo was discovered by? — Michael Faraday.
- Who discovered the 'Transformer'? — Michael Faraday.
- The scientist who gave the laws of Electrolysis? — M. Faraday.
- Who invented the Hydrogen Bomb? — Edward Teller.
- Who is credited with the discovery of the Electron? — J.J. Thomson.
- Who discovered the Proton? — Eugen Goldstein.
- Who discovered the Neutron (a particle that is part of the nucleus of an atom)? — James Chadwick.
- Heavy water is commonly used by which industry? — Nuclear Power Generation Plants.
- What is generally used to make an atomic bomb? — Uranium.
- Which is not a radioactive element? — Cadmium.
- Who propounded the Theory of Relativity? — By Albert Einstein.
- Who invented the Jet Engine? — By Frank Whittle.
- Indian scientist C.V. Raman presented a research paper on what, for which he received the Nobel Prize? — Theory of Musical Instruments / Scattering of Light.
- Which particles are inside the nucleus of an atom? — Protons and Neutrons.
- What is Heavy Water? — Deuterium (Deuterium Oxide).

- Are positively charged? — Alpha particles.
- Who discovered the antibiotic Penicillin? — Alexander Fleming.
- Which scientist received the Nobel Prize in 1995 for creating the Bose-Einstein Condensate? — Carl Wieman (with Eric Cornell).
- Which telescope did Isaac Newton invent in the 17th century that used a concave mirror? — Reflecting Telescope.
- Which device did John Ambrose Fleming invent in 1904 (important for electronic computers)? — Oscillation Valve (Diode Valve).
- Who discovered in 1928 that when a beam of colored light enters a liquid, a part of the light scattered by that liquid becomes of a different color? — C.V. Raman.
- Who invented the Induction Motor? — Nikola Tesla.
- Who discovered in 1820 that an electric current moves a nearby magnetic needle? — Hans Christian Oersted.
- Who invented the Power Battery in 1800 AD? — Alessandro Volta.
- Who invented the Airplane? — Orville and Wilbur Wright (Wright Brothers).
- Revolver was invented by? — Samuel Colt.
- Who received the Nobel Prize in Physics for discovering 'Energy Quanta' in 1918? — Max Karl Ernst Ludwig Planck.
- Superconductivity was discovered by (1913 Nobel Prize)? — Heike Kamerlingh Onnes.
- Who invented the Atomic Battery in 1912? — Henry Moseley.
- Who invented the Contact Lens? — Adolf Gaston Eugen Fick.
- Which scientist invented Ultra-Short Radio Waves? — J.C. Bose.
- Which scientist invented the Bifocal lens? — Benjamin Franklin.
- Who invented the Video Game Console? — Ralph H. Baer.
- Who is universally credited with the invention of the Barometer in 1643? — Evangelista Torricelli.
- Which scientist invented the Electromagnet? — William Sturgeon.
- Who invented the Cyclotron in 1934? — Ernest O. Lawrence.
- Who discovered the Photoelectric Effect? — H.R. Hertz.
- Who invented the Pendulum Clock in 1656? — Christiaan Huygens.
- Who invented the Phonograph? — Thomas Edison.

- Who invented the LCD Projector? — Gene Dolgoff.
- Who invented the Induction Coil? — Nicholas Callan.
- Who invented the Gas Turbine? — John Barber.
- When was the Bar Code Scanner invented? — In 1974.
- Who invented the Kaleidoscope? — David Brewster.
- Who invented the Magic Marker? — Sidney Rosenthal.
- Who invented the Revolving Door? — Theophilus Van Kannel.
- Who invented the Ceiling Fan? — Philip Diehl.
- Who invented the Electric Chair? — Alfred P. Southwick.
- Who invented the Postage Meter? — Arthur Pitney.
- Who invented the Seismometer? — Luigi Palmieri.
- Who invented the Lawnmower? — Edwin Beard Budding.
- Who invented the Electric Tram? — Fyodor Pirotsky.
- Who invented the Post-it Note? — Arthur Fry.
- Who invented the Assembly Line? — Ransom Eli Olds.
- Who invented the 3-D Printer? — Chuck Hull.
- Who invented the Dishwasher? — Josephine Cochrane.
- Who invented the Aerosol Can? — Erik Rotheim.
- Who invented the Neon Lamp? — Georges Claude.
- Who invented Universal Standard Time? — Sandford Fleming.
- Who invented the Internet Protocol? — Vint Cerf.
- Who invented the Nuclear Reactor? — Enrico Fermi.
- Who invented the Water Turbine? — Benoît Fourneyron.
- Who invented the Airbag? — Assen Jordanoff.
- Who invented the Helicopter? — Sikorsky.
- Who invented the Telephone Exchange? — Tivadar Puskas.
- Who invented the Laser? — Gordon Gould.
- What did J.B. Dunlop invent? — Pneumatic Rubber Tire.
- Who invented LED? — Nick Holonyak.
- Who invented the Lightning Conductor in the year 1749? — Benjamin Franklin.
- Who invented Cinematography? — William Friese-Greene.
- What did Isaac Newton invent? — Reflecting Telescope.

- Who invented PlayStation? — Ken Kutaragi.
- Who invented the Steam Engine? — George Stephenson.
- Who invented the Electroscope? — Jean-Antoine Nollet.
- Who invented Tetris (Video Game)? — Alexey Pajitnov.
- Who invented the Sewing Machine? — Isaac Singer.
- Who invented the Electric Stove? — Lloyd Groff Copeman.
- Who invented the Piano? — Bartolomeo Cristofori.
- Who invented Television? — J.L. Baird.
- Who invented Dynamite? — Alfred Nobel.
- Who invented the Christmas Card? — Sir Henry Cole.
- Who invented the Vacuum Flask? — James Dewar.
- Who discovered Catalysis? — Berzelius.
- Who made the first Laser? — Theodore H. Maiman.
- Who invented the Lightning Conductor? — Benjamin Franklin.
- Was invented by Thomas Edison? — Electric Bulb.
- Who invented the Stethoscope? — Rene Laennec.
- Who invented the Power Loom? — Edmund Cartwright.
- What did Walter Hunt invent? — Safety Pin.
- Who invented the Cotton Gin? — Eli Whitney.
- Who invented Kevlar? — Stephanie Louise Kwolek.
- Name the physicist who is credited with the discovery of the Neutron (1932)? — James Chadwick.
- Who invented the Rocket? — Robert Goddard.
- Who invented FM Radio? — Edwin Howard Armstrong.
- Who discovered Microwaves? — Percy Spencer.
- Who invented the Fountain Pen? — Lewis Edson Waterman.
- Who invented the Barometer? — Evangelista Torricelli.
- Who is the father/discoverer of Animation? — Émile Cohl.
- Who invented the Compound Microscope? — Zacharias Janssen.
- Who invented Television? — J.L. Baird.
- Who invented the Ammeter? — Friedrich Drexler.
- Who invented the Telephone? — Graham Bell.

- Who invented the Transformer? — Michael Faraday.
- Who invented the Microwave Oven? — Percy Spencer.
- Who invented the Battery? — Alessandro Volta.
- Who invented the Vernier Caliper? — Pierre Vernier.
- Who invented the Dynamo? — Michael Faraday.
- Who invented the Diesel Engine? — Rudolf Diesel.
- Who invented the Computer? — Charles Babbage.
- Who invented the Electron Microscope? — Ernst Ruska, Bodo von Borries, and Max Knoll.
- Who invented the Thermometer? — Daniel Gabriel Fahrenheit.

Space, Measurement & Miscellaneous Physics

- In Physics, what is the Hubble Constant? — Unit of measurement that describes the expansion of the universe.
- Which energy is NOT generated from the ocean? — Geothermal energy.
- Sudden drop in Barometer reading implies? — Storm.
- The physical quantity for the expression Arc/Radius is? — Plane Angle.
- Is not a vegetable fiber? — Nylon.
- The ratio of energy flow at different points in a food chain is known as? — Ecological Efficiency.
- Which type of lens is always used to obtain a smaller image? — Concave lens.
- The average duration of time taken by the newly developed world's smallest X-ray laser pulse, which will allow us to observe electrons in slow motion? — 43 Attoseconds.
- Name of India's first human mission in space, which aims to demonstrate indigenous capability to launch human spaceflight missions to Low Earth Orbit (LEO)? — Gaganyaan.
- When did India launch the first experimental satellite? — In the year 1975 (Aryabhata).
- In 2021, the Indian Space Research Organisation (ISRO) has signed an agreement with the space agency of which country for cooperation on the country's first human space mission Gaganyaan? — France (CNES).
- The capacity of a body to do work is called? — Energy.

- The light ray used in a pathological laboratory inside a test tube for the elimination of Bacteria is named? — Ultraviolet radiation.
- Example of Class II lever is? — Wheelbarrow.
- The wave nature of electrons was first experimentally proven by? — Davisson and Germer experiment.
- The name of the particle 'Boson' is associated with which name? — S.N. Bose.
- The speed of tape in a tape recorder is — 4.76 cm/second.
- The largest share in electricity generation is of — Thermal electricity.
- Why does 'Free Play' occur in steering? — Due to the steering being loose.
- Used in CT Scanning — X-ray.
- Lemon squeezing machine is — Second class lever.
- Stephen Hawking is a — Scientist.
- Besides being a scientist, he was also a great industrialist who established many industries in India — Vikram Sarabhai.
- While climbing mountains, air pressure — Decreases.
- The electronic configuration of an element is 2, 8, 4. The name of this element is — Silicon.
- Which metal is used in electroplating of cans for storing food? — Sn (Tin).
- What was invented in the 1890s by Italian physicist Guglielmo Marconi? — Radio.
- Invented in Britain by G. Bradshaw in 1919 — Scooter.
- Invented by American scientist Elisha Graves Otis in 1852 AD — Lift (Elevator).
- Ammonium Sulfate contains Nitrogen — 20%.
- Not required for Radiation — Medium.
- The common name of Higgs Boson particle is — God Particle.
- Water remains cool in an earthen pot — Due to evaporation.
- Laser is a device by which is generated — Stimulated radiation.
- Automatic wristwatches get energy — From various movements of our hand.
- When there is a half-moon, the angle between Sun, Earth, and Moon is — 90°.
- It is harder to walk on ice than on the road because — Friction is less in ice than on the road.
- Television signals cannot be received after a certain distance because — The surface of the Earth is curved.

- If a ship enters the sea from a river then — It will rise.
- Does not denature protein — Infrared rays.
- A person stands on a sensitive scale. If he takes a deep breath in, the scale reading will — Increase (because he will take in extra air).
- The largest share in energy production in India is — Thermal energy.
- The terminals of a simple torch cell are made of — Zinc-Carbon.
- You are traveling in a car in a thunderstorm, to protect yourself from lightning you will choose — To stay in the car.
- X-rays that can penetrate matter for a long distance are called — Hard X-rays.
- The rain-proof property of materials of 'Rain Proof' coats and tents is due to — Its Surface Tension.
- Fish survive in a frozen lake because — Fish go to the bottom of the lake where water is at 4°C.
- Television signals cannot be received after a certain distance because — The surface of the earth is curved.
- Is NOT perfectly inelastic — Kicking a football by a player.
- On increasing the temperature, the resistance of semiconductors — Increases (Note: This is written in the source, although scientifically, the resistance of semiconductors *decreases* as temperature increases. Please choose options carefully in exams, here the answer is given as per source).
- At zero Kelvin temperature, Silicon is a — Insulator.
- Which law states that under similar conditions of temperature and pressure, the rate of diffusion of gases is inversely proportional to the square root of their density — Graham's Law.
- In which year was the Doppler effect discovered by Austrian scientist Christian Doppler? — 1842.
- The following law states that the total energy per unit mass of a flowing fluid at any point in the subsurface is the sum of kinetic, potential, and fluid-pressure energy and is equal to a constant value — Daniel Bernoulli.
- What is promoted across the country to save energy — Light Emitting Diode Lamps (LED).

- Which leading scientist used radio waves to transmit signals over distances of several kilometers in the 1890s — Guglielmo Marconi.
- Henry Cavendish described the measurement of the gravitational constant along with the mass and density of the Earth — June 1798.
- The main energy source for the Earth's internal heat engine is — Radiogenic heat and Primordial heat.
- Famous physicist Victor Franz Hess is famous for the following discovery — Cosmic Radiation.
- Radiation belts are giant belts of highly energetic charged particles trapped magnetically that surround the Earth — Van Allen.
- A wooden bat can be classified as which object — Opaque.
- English physicist James Prescott Joule outlined the basis of — Principle of Conservation.
- The daily morning task is a part of quantum mechanics — Ringing of the alarm clock.
- The property of metals to produce a ringing sound when struck by a hard object is known as — Sonority.
- Awarded the Nobel Prize in Physics for 'his work on the discontinuous structure of matter' in 1926 — Jean Baptiste Perrin.
- Name the high-frequency radiation produced in nuclear reactions and emitted by radioactive nuclei, which is used in medicine to destroy cancer cells — Gamma rays.
- If a glass bottle is filled completely with water, and sealed with a lid and placed in a freezer, what will happen — Eventually ice will break the glass bottle.
- What is the name of the phenomenon in physics and astronomy that involves the splitting of a spectral line into two or more components of slightly different frequency when a light source is placed in a magnetic field — Zeeman Effect.
- Supercooling implies cooling of liquid to which point — Below freezing point.
- Alcohol is more volatile than water because — Its boiling point is less than water.
- The best adsorbent is — Activated Coconut Charcoal.
- If the vector sum of torques acting on a system is zero, then that object is — In Rotational Equilibrium.
- The theoretically possible maximum mass for a stable white dwarf star is called — Chandrasekhar Limit.

- Green plants convert solar energy into — Chemical energy.
- Which famous scientist raised the question, 'If time travel is possible, then where are the tourists from the future' — Stephen Hawking.
- This is the statement of which physicist, "Physics is always a gamble" — David Gross.
- Stars twinkle, but planets do not because — They are closer to the Earth, so more light is received from them and even if there are minor changes in intensity, they are not visible.
- The component used for tuning the radio is basically a ..... — Condenser/Capacitor.
- A group of solar cells connected together in a definite pattern is called — Solar Cell Panel.
- Which compound is used in prickly heat powder to prevent excessive sweating — Boric acid.
- Transistors are most likely to be found in — Audio equipment.
- If an electron and a photon have the same wavelength, then they will have the same — Linear momentum.
- In IT, to increase 'bandwidth', the technique used is — Memory Interleaving.
- A green leaf plant placed in red light will appear — Black.
- A closed container in a fast-moving train contains some gas at atmospheric pressure and room temperature. The temperature of the gas will — Remain unchanged.
- Gliders rely on the following energy for their flight — Wind energy.
- At what point does the cancellation of the bangle occur — Cancellation point (Note: Literal translation, context likely refers to wave cancellation or specific mechanical term).
- A body in equilibrium — Can move with constant velocity.
- Due to what reason is there a variation in the boiling point of liquids — Variation in pressure.
- Why does a person standing on the platform of a railway station feel a pull towards a rapidly approaching train — Difference in pressure due to fast-moving air between them (Bernoulli's principle).
- An example of the motion of a train is — Translatory motion.
- During the event of thunder and lightning in the sky — Lightning is seen before thunder is heard.

- The term 'Askaryan effect' is related to which functional area of science — Physics.
- Responsible for the rotation of the plane of oscillation of Foucault's pendulum — Coriolis force.
- The word ..... is used as a prefix to describe constant scientific terms — Iso.
- Propounded the theory of relativity — Albert Einstein.
- Heat is a component of the environment — Abiotic component.
- Is not an insulator — Mercury.
- ..... is a pair of forces that are equal in magnitude, directed in opposite directions, and displaced by a straight distance or moments — Couple.
- Is not an electrical insulator — Gold.
- Is not an essential part of a compound microscope in a school laboratory — Camera.
- Apiculture is related to — Beekeeping.
- Silviculture is related to — Mulberry tree (Note: Generally Silviculture is forestry, but the source says Mulberry).
- Viticulture is related to — Grape cultivation.
- Pisciculture is related to — Fish farming.

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